PROJECT	DESIGNATION
1593276	1593276
CONTRACT	BRIDGE FILE
B-42017	(421) 39-12-01792C

STRUCTURE INFORMATION				
STRUCTURE	TYPE	SPAN AND SKEW	OVER	STATION
(421) 39-12-01792C	Steel Truss Bridge with Prestressed Concrete Box Beam End Spans	3 Spans: 30'-8½", 125'-0", 30'-8½" Skew: Square	South Fork Wildcat Creek	Sta. 43+50.00 Line "A"

INDIANA DEPARTMENT OF TRANSPORTATION

BRIDGE REHABILITATION PLANS

FOR SPANS OVER 20 FEET

ROUTE: US 421 AT: RP 126+82

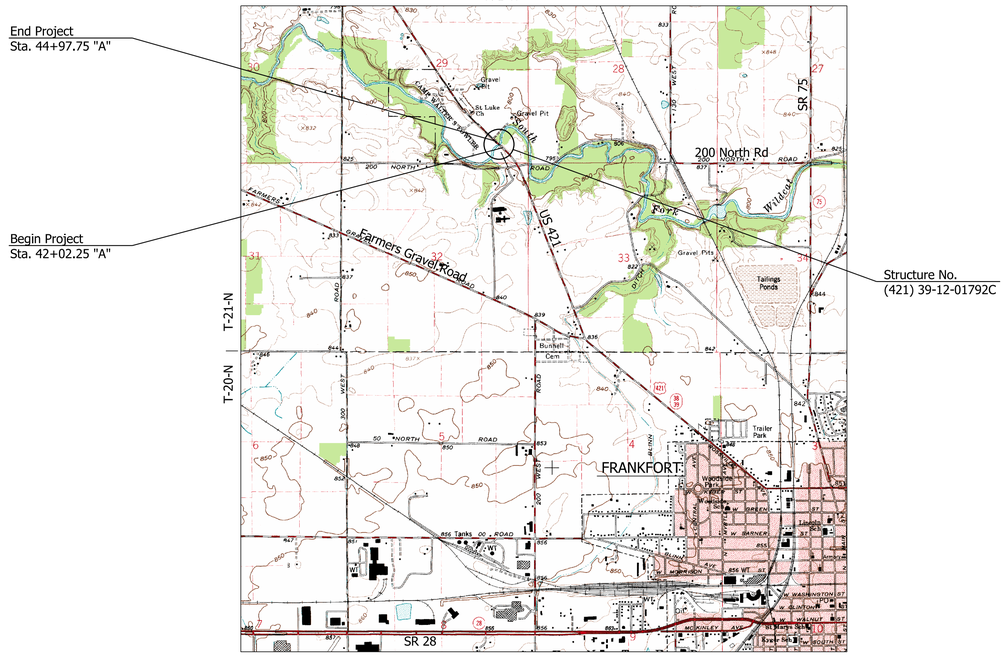
PROJECT NO. 1593276 P.E.

NO ADDITIONAL RIGHT-OF-WAY REQUIRED FOR THIS PROJECT

R/W

1593276 CONST.

Bridge Deck and End Spans Replacement on US 421 over South Fork Wildcat Creek, located 2.24 Miles south of SR 38, in Section 29, T-21-N, R-1-W, Union Township, Clinton County, Indiana

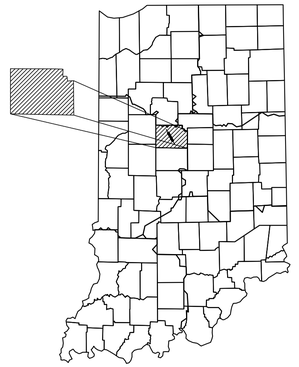


SCALE: 1" = 2000'

Clinton County

TRAFFIC DATA		
A.A.D.T.	(2020)	4,425 V.P.D.
A.A.D.T.	(2040)	6,298 V.P.D.
D.H.V		567 V.P.H.
DIRECTIONAL DISTRIBUTION		50/50 %
TRUCKS		12 % A.A.D.T.
		9 % D.H.V.

DESIGN DATA	
DESIGN SPEED	55 M.P.H.
PROJECT DESIGN CRITERIA	3R (NON-FREEWAY)
FUNCTIONAL CLASSIFICATION	MINOR ARTERIAL
RURAL/URBAN	RURAL
TERRAIN	LEVEL
ACCESS CONTROL	NONE



PROJECT LOCATION SHOWN BY
CLINTON COUNTY

LATITUDE: 40° 18' 59" N LONGITUDE: 86° 32' 48" W

BRIDGE LENGTH:	0.037	MI.
ROADWAY LENGTH:	0.020	MI.
TOTAL LENGTH:	0.057	MI.
MAX. GRADE:	0.0	%

gai consultants

Indianapolis: 201 North Illinois Street, Suite 1700, Indianapolis, IN 46204
Fishers: 9998 Crosspoint Boulevard, Suite 110, Indianapolis, IN 46256

INDIANA DEPARTMENT OF TRANSPORTATION
STANDARD SPECIFICATIONS DATED 2020
TO BE USED WITH THESE PLANS.

PLANS PREPARED BY: GAI Consultants, Inc. (317) 436-9150
PHONE NUMBER

CERTIFIED BY: _____ DATE

APPROVED FOR LETTING: _____ DATE
INDIANA DEPARTMENT OF TRANSPORTATION

BRIDGE FILE
(421) 39-12-01792 C
DESIGNATION
1593276

SHEETS		
1	of	30
CONTRACT	PROJECT	
B-42017	1593276	

Plot: 2/14/2020 10:30:39 AM By: thomatm Pen: Transportation.tbl

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UTILITIES

Communications

AT&T - Distribution
116 E Taylor St
Kokomo, IN. 46901
Curtis Miller
(765) 454-4054
cm5432@att.com

Electric

Frankfort City Light & Power
1000 Washington Avenue
Frankfort, IN 46041
Michael Myers
(765) 654-4424
mmyers@fmu-in.com

REVISIONS

SHEET NO.	DATE	REVISED

GENERAL NOTES

Roadway stationing is based off of Bridge Contract No. 2097 original construction plans, dated 1940.

All disturbed areas shall be mulch seeded, unless noted otherwise.

INDEX

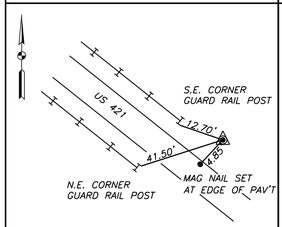
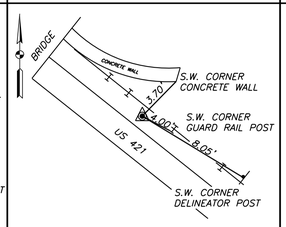
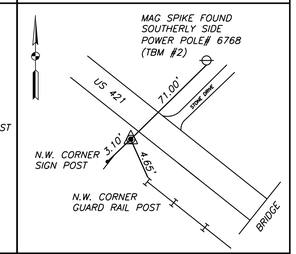
SHEET NO.	DRAWINGS INDEX
1	TITLE
2	INDEX
3	MAINTENANCE OF TRAFFIC
4	EROSION CONTROL DETAILS
5	LAYOUT
6-7	GENERAL PLAN
8-9	REMOVAL DETAILS
10-11	ABUTMENT NO. 1 DETAILS DETAILS
12-13	PIER NO. 2 & NO. 3 DETAILS
14-15	ABUTMENT NO. 4 DETAILS DETAILS
16-19	STRUCTURAL STEEL DETAILS
20	FRAMING PLAN
21	BEAM DETAILS
22-25	SUPERSTRUCTURE DETAILS
26	DRAINAGE DETAILS
27	CONCRETE RAILING DETAILS
28	APPROACH SLAB DETAILS
29	BRIDGE SUMMARY TABLES
30	ROADWAY SUMMARY TABLES

BENCH MARKS:

CLIN C-22 - Located on US 421/SR 39/SR 38, 2.2 miles southeast of SR 38 north split on top of Pier Cap #3 (northwest corner) of bridge over South Fork Wildcat Creek. Elevation 792.60

TBM #1 - Top of mag spike set in Westerly Side of power pole #6765 on N. Side of US 421 at Northwest end of bridge. Elevation: 789.91

TBM #2 - Top of mag spike set in Westerly Side of power pole #6768 on N. Side of US 421 at Southeast end of bridge. Elevation: 792.07

REFERENCE POINT #1	REFERENCE POINT #2	REFERENCE POINT #3
		
N. 178626.909 / E. 802438.340	N. 178727.531 / E. 802328.853	N. 178938.159 / E. 802063.401
5/8" REBAR W/"TRAV PT GAI 0007"CAP SET	MAG SPIKE W/"GAI 0007" BRASS DISK SET	5/8" REBAR W/"TRAV PT GAI 0007"CAP SET

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER DATE

DESIGNED: NRT DRAWN: TMT

CHECKED: KMP CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

INDEX

SCALE	BRIDGE FILE
NONE	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	2 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:30:54 AM By: thomatm Pen: Transportation.tbl

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SIGN LEGEND:

(A) Advance Turn Detour Route Marker Assembly
Detour (XM4-8)
Cardinal Direction (North, M3-1(S)) or (South, M3-3(S))
US Route (421) (M1-4)
Advance Turn Arrow (M5-1 or M5-2) (L or R)

(B) Directional Detour Route Marker Assembly
Detour (XM4-8)
Cardinal Direction (North, M3-1(S)) or (South, M3-3(S))
US Route (421) (M1-4)
Directional Arrow (M6-1 or M6-2) (L or R)

(C) Confirming Detour Route Marker Assembly
Detour (XM4-8)
Cardinal Direction (North, M3-1(S)) or (South, M3-3(S))
US Route (421) (M1-4)
Directional Arrow (M6-3)

(D) End Detour Route Marker Assembly
End Detour (XM4-8a)
Cardinal Direction (North, M3-1(S)) or (South, M3-3(S))
US Route (421) (M1-4)

(E) DETOUR AHEAD
XW20-2
36"x36"

(F) ROAD CLOSED AHEAD
XW20-3
36"x36"

(G) ROAD CLOSED 500 FT
XW20-3
36"x36"

(H) ROAD CLOSED 1000 FT
XW20-3
36"x36"

(I) ROAD CLOSED
_ MILES AHEAD
LOCAL TRAFFIC ONLY
R11-3
60"x30"

(J) ROAD CLOSED
R11-2
48"x30"

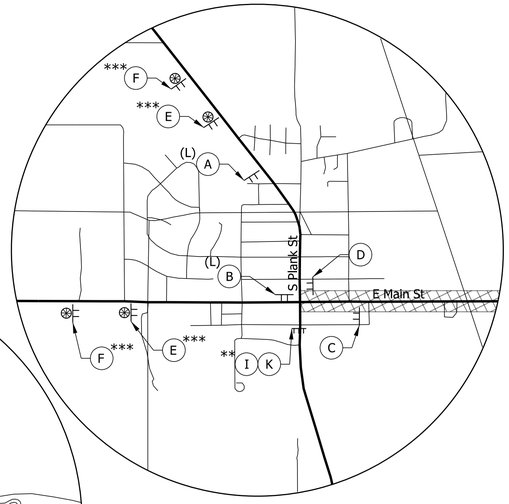
(K) DETOUR
XM4-10 (L or R)
48"x18"

(L)
R3-2
24"x24"

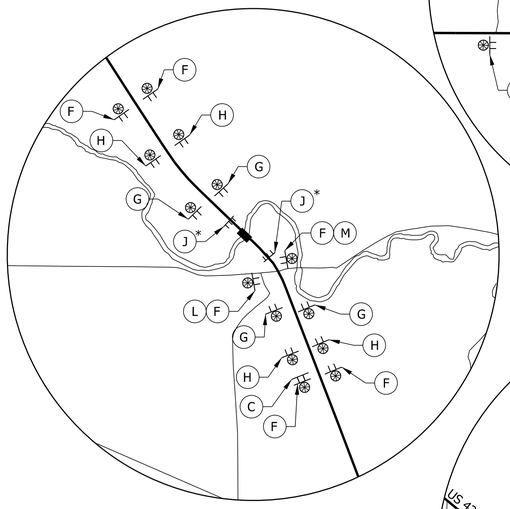
(M)
R3-1
24"x24"

LEGEND

- Detour Route
- Construction Sign
- Low Intensity Construction Warning Light, Type A
- Type III Barricade
- Closure Area
- * 24 Lft. of Type III-A Barricade
- ** 16 Lft. of Type III-B Barricade
- *** With US 421 Placard
  Cardinal Direction (North, M3-1(S)) or (South, M3-3(S))
  US Route (421) (M1-4)
- **** With SR 38 Placard
  Cardinal Direction (East, M3-2(S)) or (West, M3-4(S))
  SR Route (38) (M1-4)



DETAIL 'C'
Not to Scale



DETAIL 'B'
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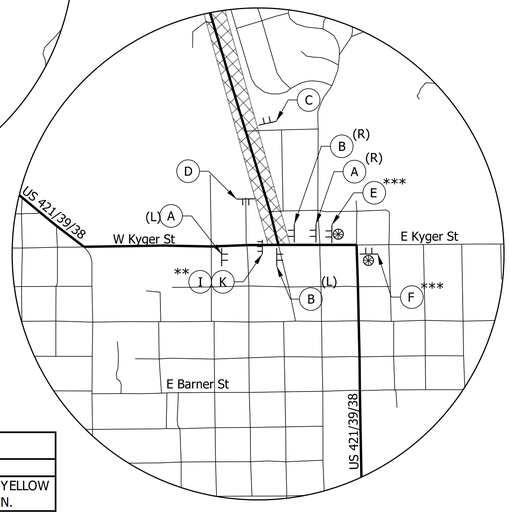

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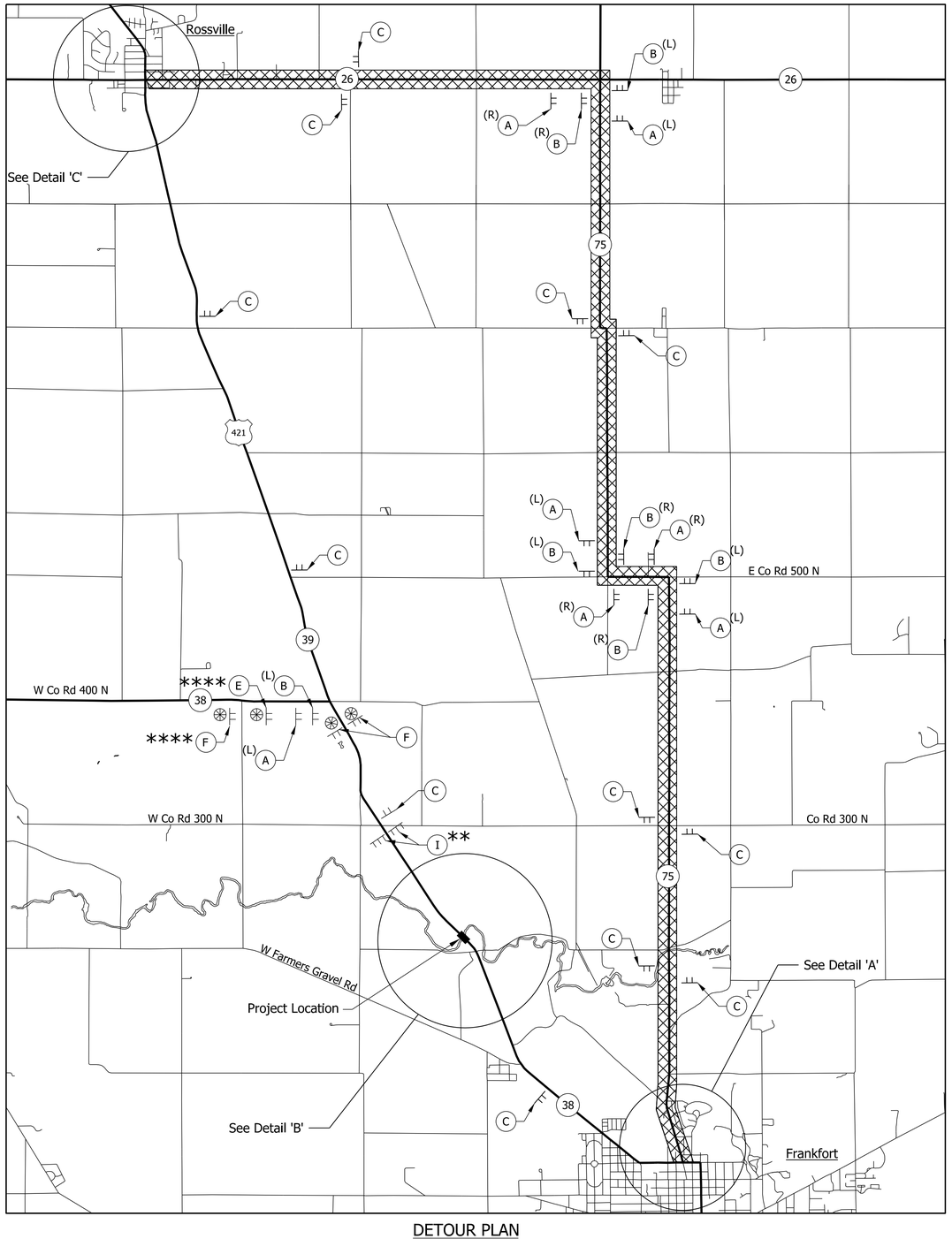

DETOUR PLAN
SCALE: 1" = 5,000'

NOTES:

1. Refer to INDOT Standard Drawing E 801-TCDT-01 for placement of signs and devices in a rural detour and E 801-TCDT-04 for Detour Route Marker Assembly Details.
2. Two - "Route Number Closed On or After ___" Signs (XG20-5) to be placed as directed by the engineer prior to construction.
3. Upon completion of the project restripe the double yellow centerline and solid white edge lines.
4. Install R11-2 Road Closure sign assemblies on Type III-A barricades. Install R11-3 Road Closure sign assemblies on Type III-B barricades.
5. Conduct flagging during paving operations in accordance with E 801-TCTC-05.

PAVEMENT MARKINGS TABLE

LOCATION	LINE, MULTI-COMPONENT		
	SOLID YELLOW 4 IN.	SOLID WHITE 4 IN.	BROKEN YELLOW 4 IN.
UNIT	LFT	LFT	LFT
TOTALS	390	780	100

MAINTENANCE OF TRAFFIC QUANTITIES

ITEM	UNITS	QUANTITY TOTAL
CONSTRUCTION SIGN, A	EACH	21
CONSTRUCTION SIGN, B	EACH	10
ROAD CLOSURE SIGN ASSEMBLY	EACH	5
DETOUR ROUTE MARKER ASSEMBLY	EACH	35
BARRICADE, III-A	LFT	48
BARRICADE, III-B	LFT	36

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER DATE

DESIGNED: NRT DRAWN: TMT

CHECKED: KMP CHECKED: NRT

INDIANA
DEPARTMENT OF TRANSPORTATION

MAINTENANCE OF TRAFFIC

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
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CONTRACT	PROJECT
B-42017	1593276

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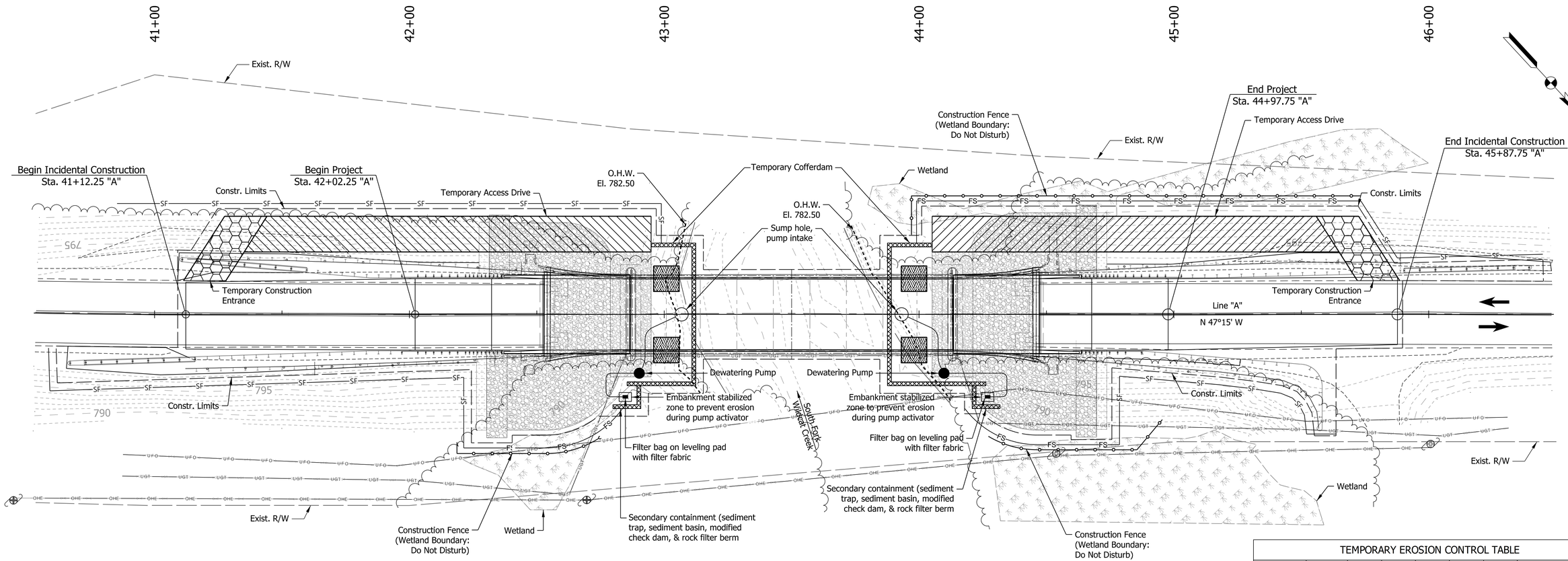
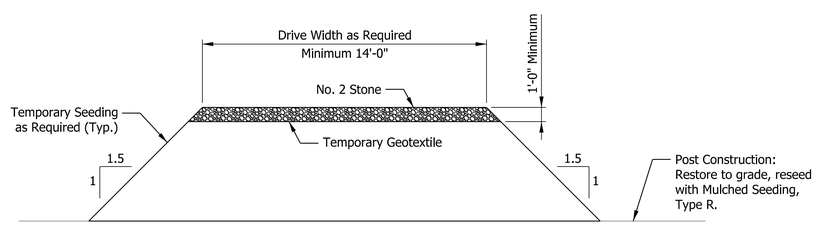
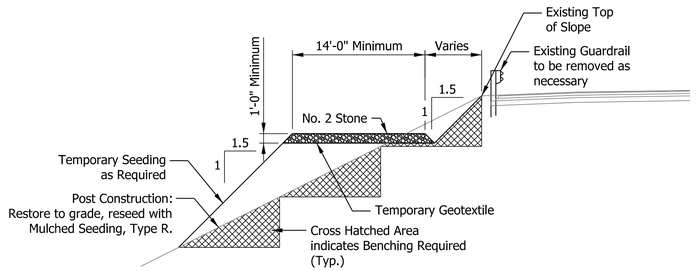


PLAN
Scale: 1" = 20'-0"



TYPICAL SECTION - TEMPORARY ACCESS DRIVE
No Scale



TYPICAL SECTION - TEMPORARY ACCESS DRIVE ADJACENT TO ROADWAY
No Scale

TEMPORARY EROSION CONTROL TABLE

STATION	LOCATION		TEMPORARY SILT FENCE	FILTER SOCK	TEMPORARY SEEDING	TEMPORARY MULCHING	FERTILIZER	REMARKS
	LEFT	RIGHT	LFT	LFT	LBS	TON	TON	
40+50		X	225					
41+00	X		230					
42+50		X		55				
44+00	X			191				
44+25		X		70				
44+75		X	125					
45+75	X		85					
TOTAL			665	316	45	0.75	0.15	

NOTES

1. Provide silt fence around the perimeter of construction and temporary staging areas.
2. Provide Filter Sock at perimeter of bridge embankment construction adjacent to channel.
3. Construct Stable Construction Entrances (Est. Qty. = 100 Tons of No. 2 Stone, 235 SYS Temporary Geotextile).
4. Provide concrete washouts as required.
5. No Causeways are permitted in the channel.
6. For additional information regarding access roads, see Special Provisions.
7. The construction access details show on this sheet were developed for permitting. If alternate methods are approved by the Engineer, the contractor shall be responsible for any modifications to design, details, permits, and any associated costs.
8. Temporary Cofferdam shall be sheet pile or other method. Materials and shape vary depending on needs and availability. Cofferdam shall be 2' minimum high and extend 1' above the ordinary high water elevation.
9. Construction fence to be placed adjacent to wetlands at perimeter of construction (See special provisions) Est. Qty. 300 Lft.

LEGEND

- SF — SILT FENCE
- FS — FILTER SOCK
- CONSTRUCTION FENCE
- ANTICIPATED LOCATION OF TEMPORARY SUPPORTS FOR JACKING AND SUPPORTING TRUSS.
- TEMPORARY ACCESS DRIVE
- TEMPORARY CONSTRUCTION ENTRANCE

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: NRT | DRAWN: TMT | CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

EROSION CONTROL PLAN

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
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CONTRACT	PROJECT
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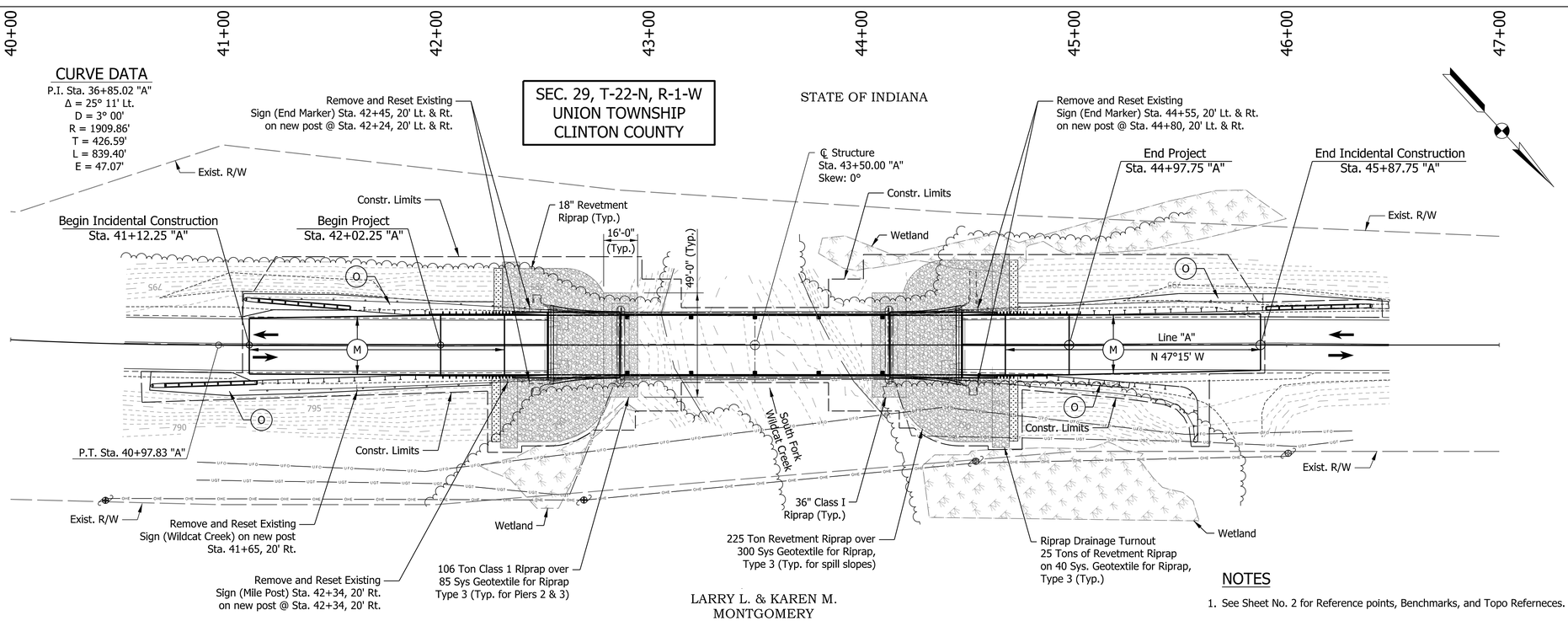


EXISTING STRUCTURES

The present structure is a steel thru truss with concrete cast-in-place girder end spans. It was built in 1941 with a 28'-0" clear roadway. The structure was rehabilitated in 1974 and 1985. The concrete end span superstructures are to be removed along with the concrete deck of the steel truss span.

EARTHWORK SUMMARY

Common Excavation	170 Cys.
Excavation Foundation, Unclassified	20 Cys.
∗ Mulched Seeding	1000 Sys.

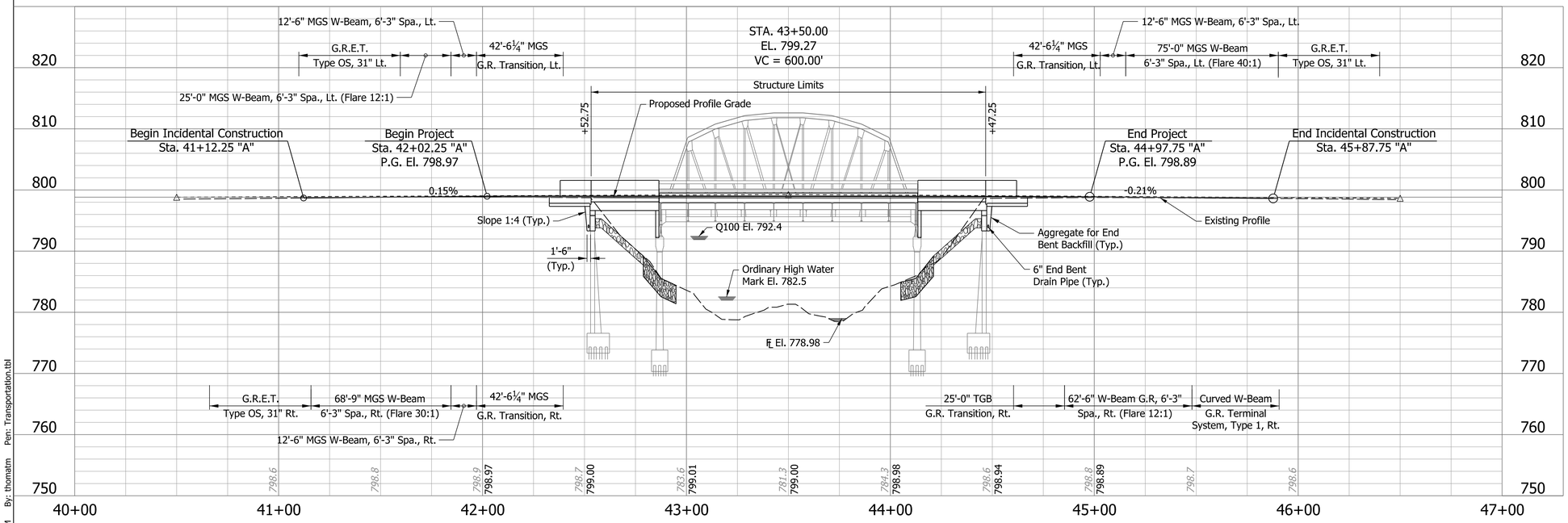
∗ Undistributed Estimated Quantity to be placed as directed by the Engineer

HYDRAULIC SCOUR DATA

Drainage Area =	76.0 sq.mi.
Design Discharge, Q_{100} =	9,050.0 cfs
Q_{100} Elevation =	792.4 ft.
Max. Velocity at Q_{100} =	7.11 ft/sec
Avg. Velocity at Q_{100} =	6.09 ft/sec
Scour Depth (Contraction) =	12.98 ft.
Scour Depth (Total) =	19.80 ft.
Low Scour Elevation =	759.18 ft.

NOTES

1. See Sheet No. 2 for Reference points, Benchmarks, and Topo Referneces.
2. Install new Delinators on new sign post at all 4 quadrants at 25' spacing. 3 Each quadrant. (12 Total)
3. All slope treatment to be Mulched Seeding, R



STEEL TRUSS BRIDGE WITH PRESTRESSED CONCRETE BOX BEAM END SPANS
3 SPANS: 31'-8½", 125'-0" & 31'-8½"
SKEW: SQUARE; 28'-0" CLEAR ROADWAY
US 421 OVER SOUTH FORK WILDCAT CREEK
CLINTON COUNTY

LEGEND
(M) See Sheet 7 for details
(O) See Sheet 7 for details

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: NRT | DRAWN: TMT | CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

LAYOUT

HORIZONTAL SCALE	BRIDGE FILE
1"=30'-0", U.N.	(421) 39-12-01792 C
VERTICAL SCALE	DESIGNATION
1"=10'-0", U.N.	1593276
	SHEETS
	5 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:23 AM By: thomatm Pen: Transportation.tbl

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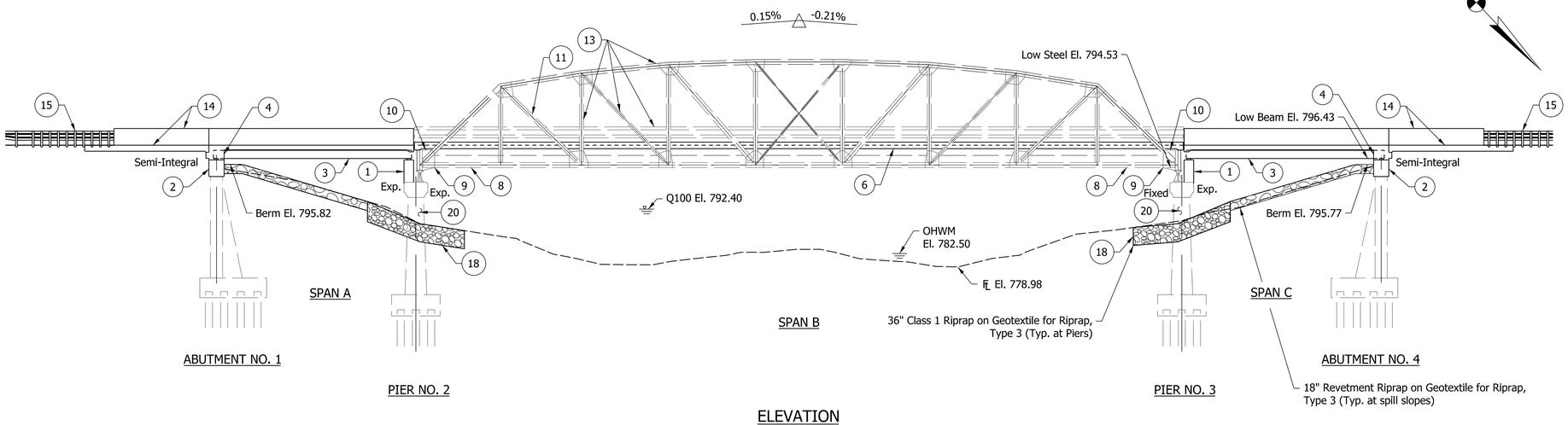
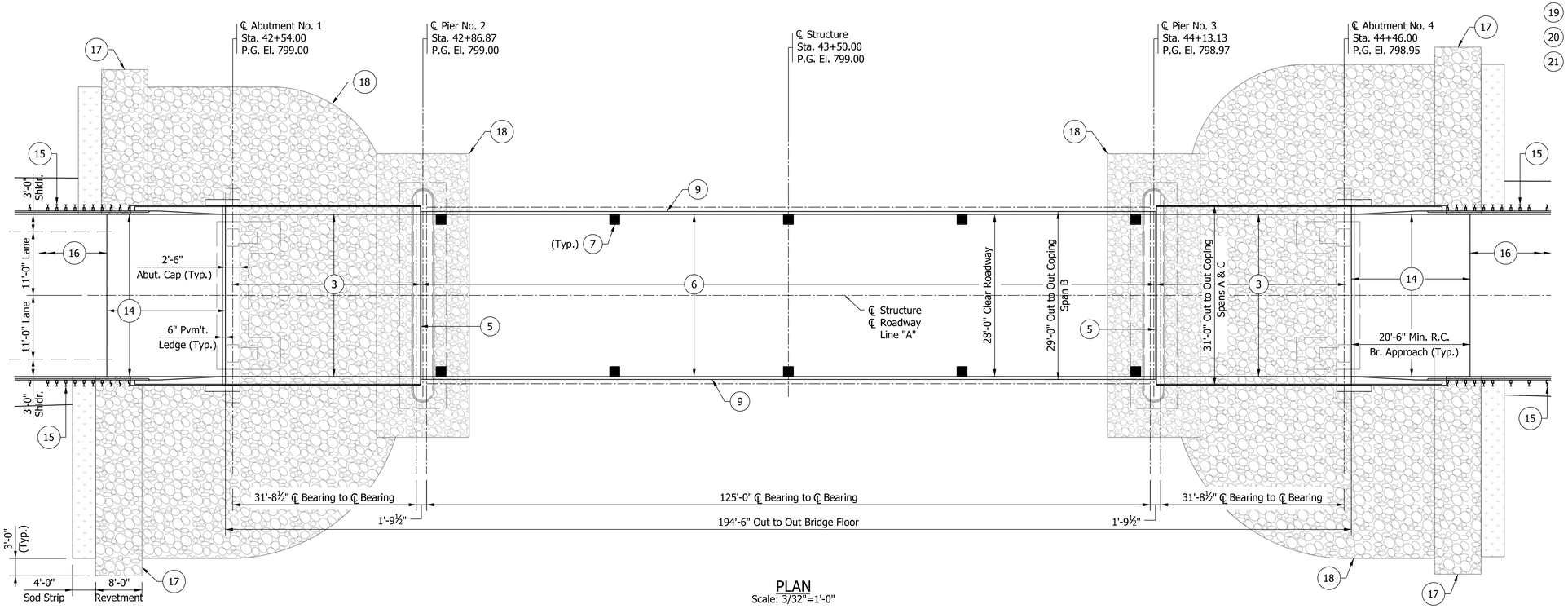
STRUCTURE BUILT TO A 600' VERTICAL CURVE

STA. 43+50.00
EL. 799.27
VC = 600.00'



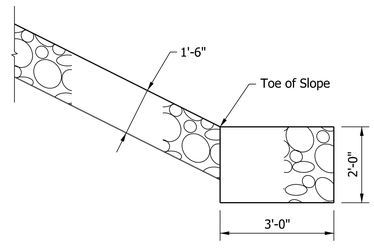
ELEVATION
Scale: 3/32"=1'-0"



PLAN
Scale: 3/32"=1'-0"

BRIDGE REHABILITATION KEY

1. Replace reinforced concrete pier pedestals for spans A & C.
2. Replace Abutment Caps 1 & 4.
3. Replace the end spans with new prestressed concrete box beam superstructures, a new reinforced concrete deck, and new Type FC Concrete Railing.
4. Construct new semi-integral end bent diaphragms at abutments 1 & 4.
5. Install new pre-compressed foam joints.
6. Replace the existing reinforced concrete deck on the steel thru-truss main span.
7. Install new bridge deck drains, Type SQ-A.
8. Temporarily Jack and support Truss.
9. Temporarily Jack and support floor beam.
10. Remove and replace Gusset Plates.
11. Construct cover plate repair at Diagonal UB-L7 Y.
12. Replace deteriorated members or portions of members as directed by engineer following structural inspection during construction. (See Special Provisions)
13. Clean and paint the existing steel thru-truss, bearings and the attached existing metal guardrail.
14. Construct new Reinforced Concrete Bridge Approaches with Type TFC Concrete Bridge Railing Transitions.
15. Replace existing guardrail at all four bridge corners.
16. Mill and Resurface existing asphalt pavement as necessary to tie back into existing.
17. Construct Riprap Turnouts at the ends of the TFC Concrete Bridge Railing Transitions.
18. Add channel scour protection.
19. Surface Seal the deck, bridge rail, copings, approach slabs and bridge rail transitions.
20. Patch concrete structures as directed by the Engineer. (Approx. 100 SFT, each pier)
21. Remove and reset sign post and delineators at approaches. (See Layout for locations)



RIPRAP REVETMENT TOE SECTION
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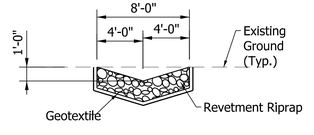

RIPRAP TURNOUT TYPICAL SECTION
Not to Scale

STEEL TRUSS BRIDGE WITH PRESTRESSED CONCRETE BOX BEAM END SPANS
3 SPANS: 31'-8½", 125'-0" & 31'-8½"
SKEW: SQUARE; 28'-0" CLEAR ROADWAY
US 421 OVER SOUTH FORK WILDCAT CREEK
CLINTON COUNTY

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RECOMMENDED FOR APPROVAL _____ - DESIGN ENGINEER DATE

DESIGNED: NRT | DRAWN: TMT
CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

GENERAL PLAN

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	6 of 29
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 8:46:09 AM By: thomatm Pen: Transportation.tbl

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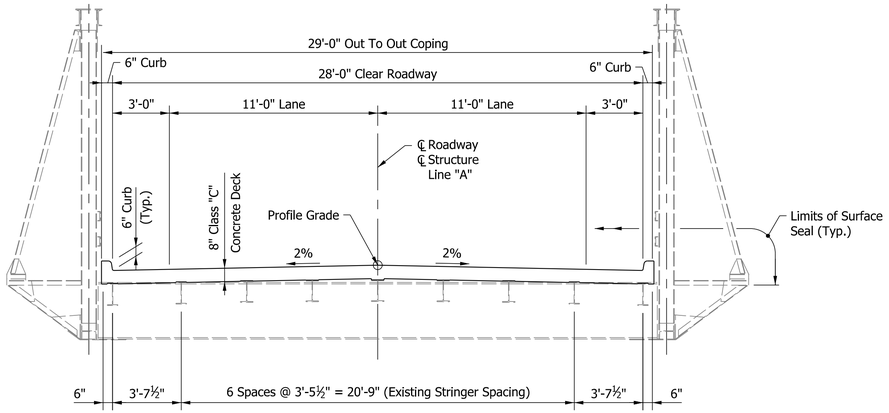
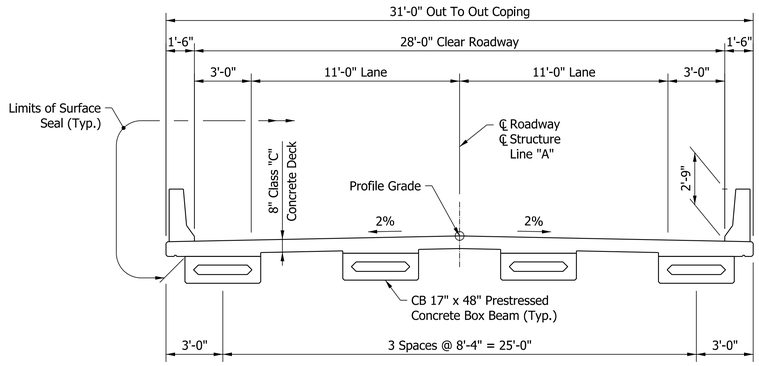
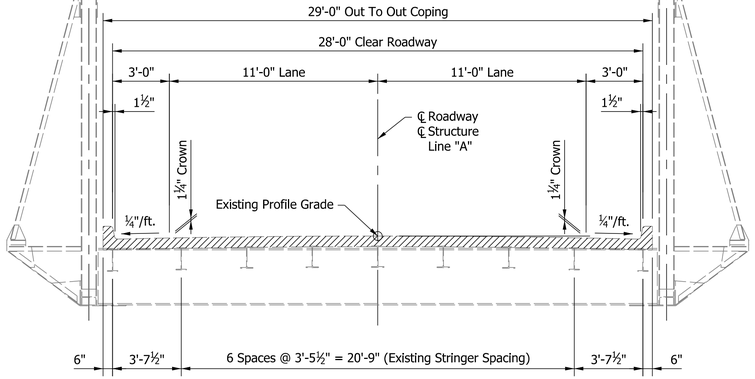


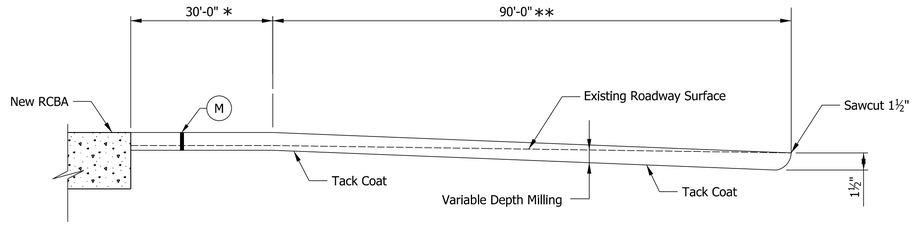
TYPICAL PROPOSED SECTION
SPAN "B"
Scale:¼" = 1'-0"



TYPICAL PROPOSED SECTION
SPANS "A" & "C"
Scale:¼" = 1'-0"

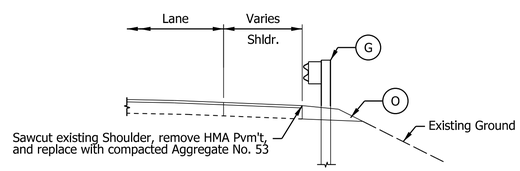


TYPICAL REMOVAL SECTION
SPAN "B"
Scale:¼" = 1'-0"



∗ - Wedge to be a continuation of bridge profile.
∗∗ - Incidental Construction.

HMA WEDGE AND LEVEL DETAIL
Not to Scale



GUARDRAIL SECTION
Appicable where new guardrail posts fall within existing shoulder footprint

GENERAL NOTES

Reinforcing steel cover shall be 2 1/2"in top and 1" minimum in bottom of floor slab, and 2" in all other parts, unless noted otherwise.

Plans for the existing structure are on file in the central office of the Indiana Department of Transportation as bridge file (421)39-12-01792, (421)39-12-01792 A, and (421)39-12-01792 B and are available upon request.

Where new work is to be fitted to old work, the contractor shall check all dimensions and conditions in the field, report any errors or discrepancies to the engineer and assume responsibilty for their correctness and the fit of the new part to the old.

DESIGN DATA

New Superstructure and deck designed for HL-93 loading in accordance with AASHTO LRFD Bridge Design Specifications Eighth Edition and interims through 2019.

DEAD LOAD

Actual weight plus 35 psf (composite) for future wearing surface and 15 (non-composite) for permanent metal deck forms (New Spans Only).

FLOOR SLAB

Designed with a 7 1/2" structural depth plus a 1/2" sacrificial wearing surface.

DESIGN STRESSES

CONCRETE

Class "A" Concrete:	f'c = 3,500 psi
Class "C" Concrete:	f'c = 4,000 psi

REINFORCING STEEL

Grade 60 Fy = 60,000 psi

CONSTRUCTION LOADING

The exterior beam has been checked for strength, deflection, and overturning using the construction loads shown below. Cantilever overhang brackets were assumed for support of the deck overhang past the edge of the exterior beam. Finishing machine was assumed to be supported 6 in. outside the vertical coping form. The top overhang brackets were assumed to be located 6 in. past the edge of the vertical coping form. The bottom of overhang brackets were assumed to be braced against the Box Beam Superstructure (New Spans Only).

DECK FALSEWORK LOADS

Designed for 15 lb/ft2 for permanent metal stay-in-place deck forms, removable deck forms, and 2-ft exterior walkways (New Spans Only).

CONSTRUCTION LIVE LOAD

Designed for 20 lb/ft2 extending 2 ft past the edge of coping and 75 lb/ft vertical force applied at a distance of 6 in. outside the face of coping over a 30-ft length of the deck centered with the finishing machine (New Spans Only).

FINISHING MACHINE LOAD

4500 lb distributed over 10 ft along the coping (New Spans Only).

WIND LOAD

Designed for 70 mph horizontal wind loading in accordance with LRFD 3.8.1 (New Spans Only).

STEEL TRUSS BRIDGE WITH PRESTRESSED CONCRETE BOX BEAM END SPANS
3 SPANS: 31'-8½", 125'-0" & 31'-8½"
SKEW: SQUARE; 28'-0" CLEAR ROADWAY
US 421 OVER SOUTH FORK WILDCAT CREEK
CLINTON COUNTY

LEGEND

- (G) MGS Guardrail
- (M) Transition Milling and 165#/Sys. QC/QA-HMA, 3, 64, Surface, 9.5 mm
- (O) Compacted Aggregate No. 53 for Shoulders

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RECOMMENDED FOR APPROVAL _____ - _____ DESIGN ENGINEER DATE

DESIGNED: NRT DRAWN: TMT

CHECKED: KMP CHECKED: KMP

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GENERAL PLAN

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
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CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:27 AM By: thomatm Pen: Transportation.tbl

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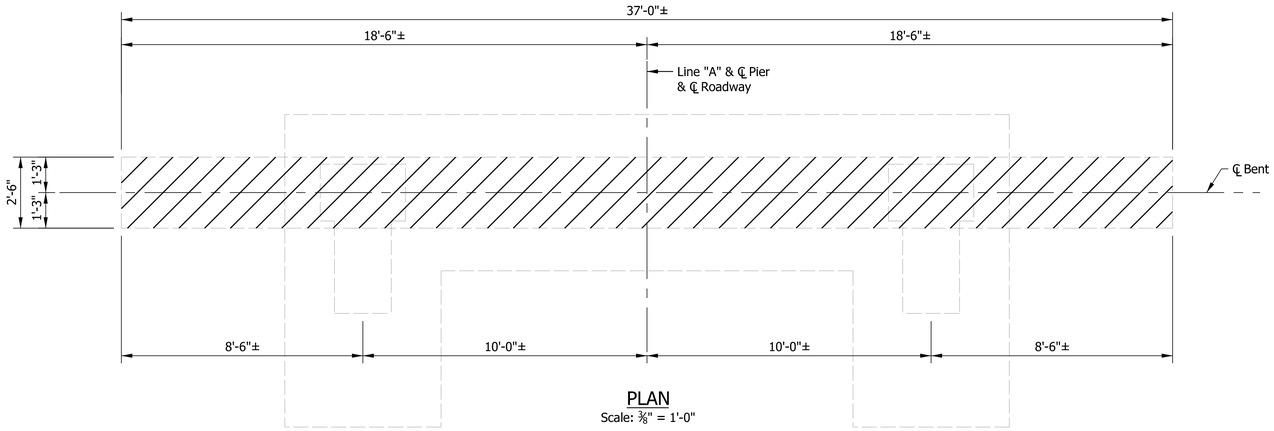
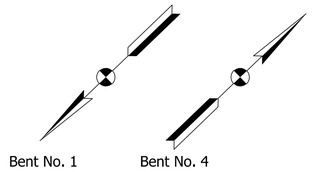
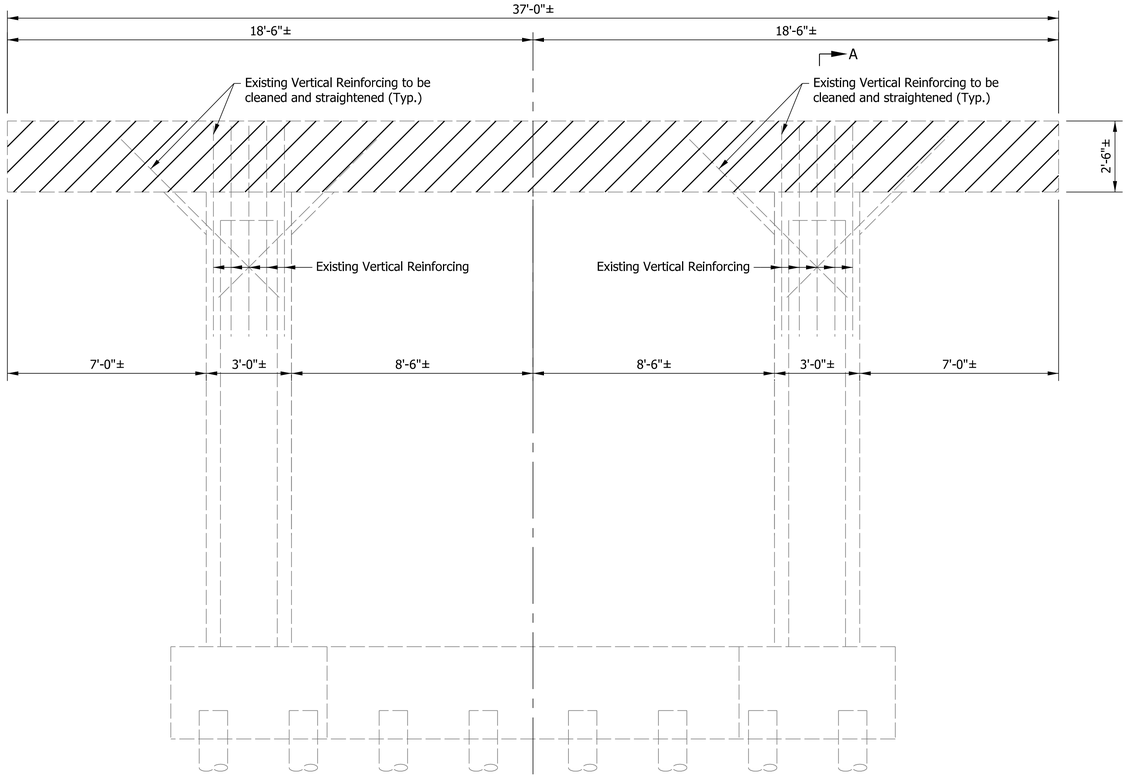


PLAN
Scale: ⅜" = 1'-0"

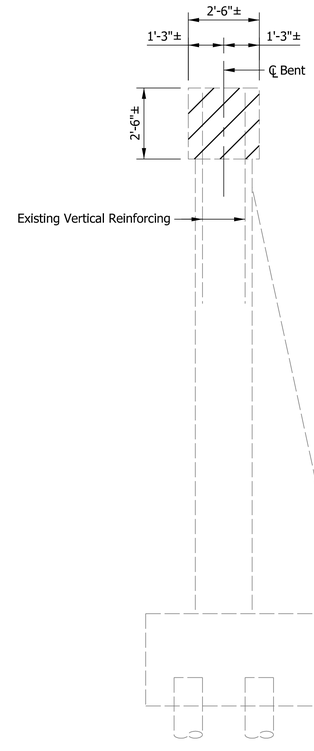


Indicates limits of Concrete Removal



ELEVATION
Scale: ⅜" = 1'-0"

A



SECTION A-A
Scale: ⅜" = 1'-0"

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RECOMMENDED FOR APPROVAL _____ - DESIGN ENGINEER DATE

DESIGNED: NRT | DRAWN: TMT

CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

BENTS NO. 1 & NO. 4
REMOVAL DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	8 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:29 AM By: thomatm Pen: Transportation.tbl

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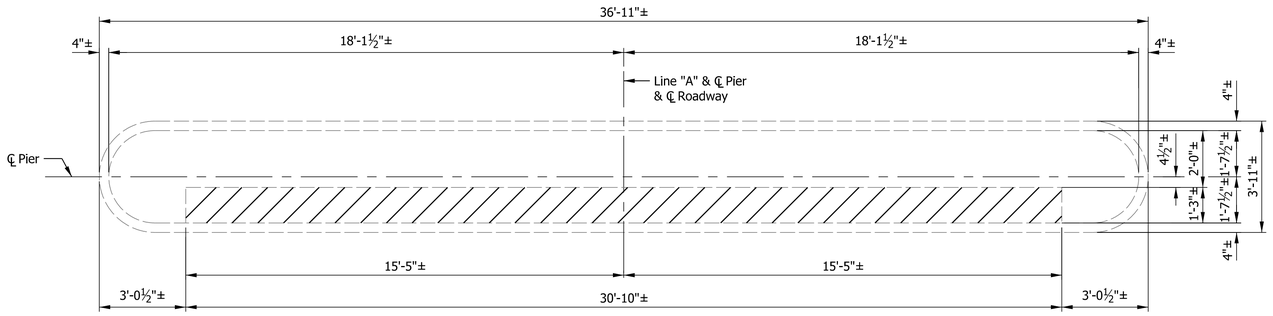
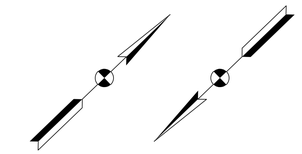
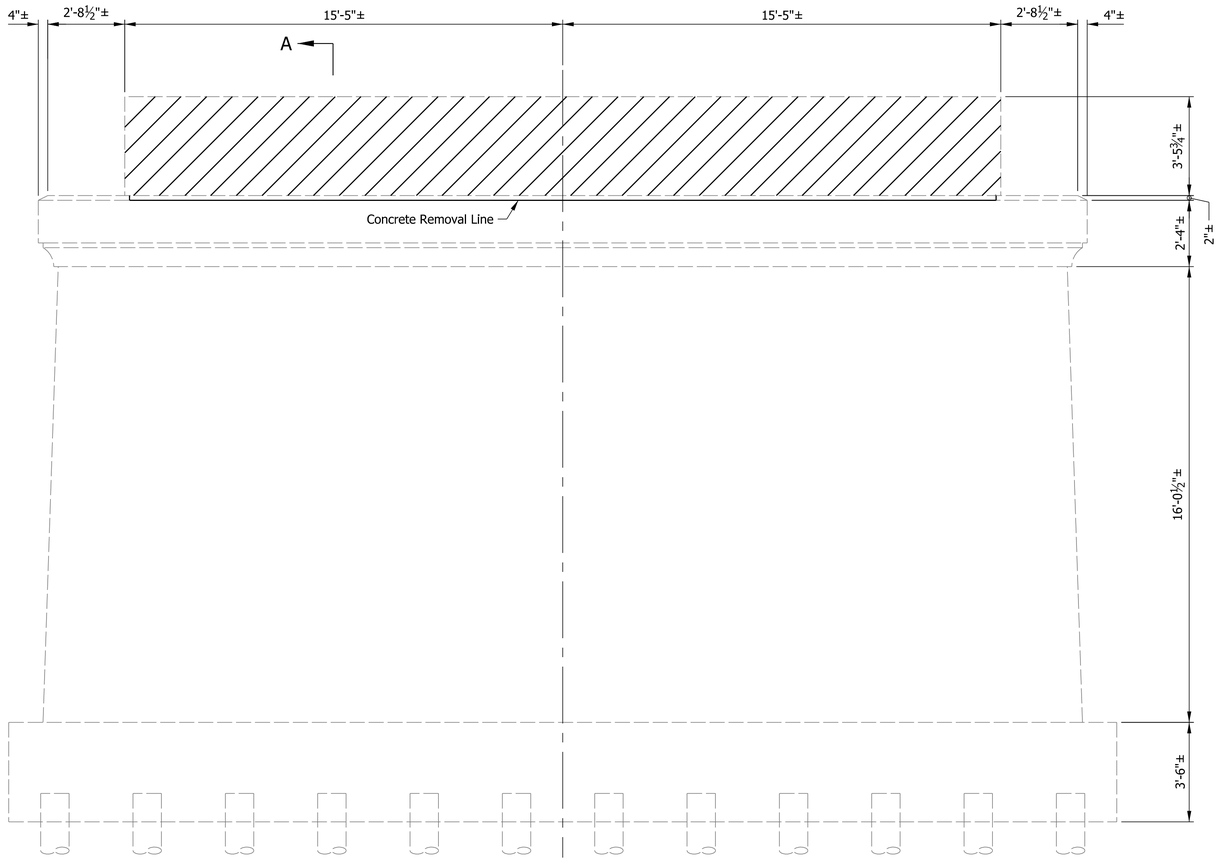


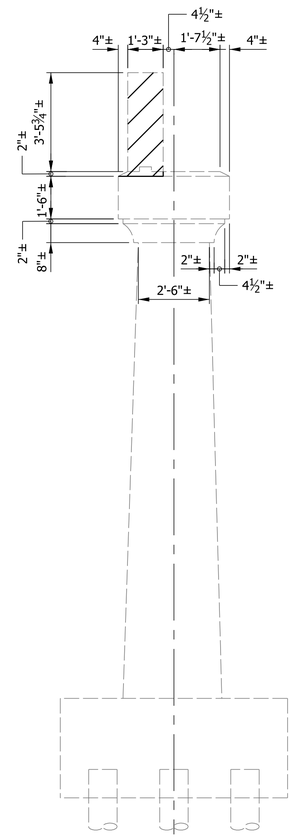
PLAN
Scale: ⅜" = 1'-0"



Indicates limits of Concrete Removal



ELEVATION
Scale: ⅜" = 1'-0"



SECTION A-A
Scale: ⅜" = 1'-0"

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RECOMMENDED FOR APPROVAL _____ - DESIGN ENGINEER DATE

DESIGNED: NRT | DRAWN: TMT

CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

PIERS NO. 2 & NO. 3
REMOVAL DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
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CONTRACT	PROJECT
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Plot: 2/14/2020 10:31:31 AM By: thomatm Pen: Transportation.tbl

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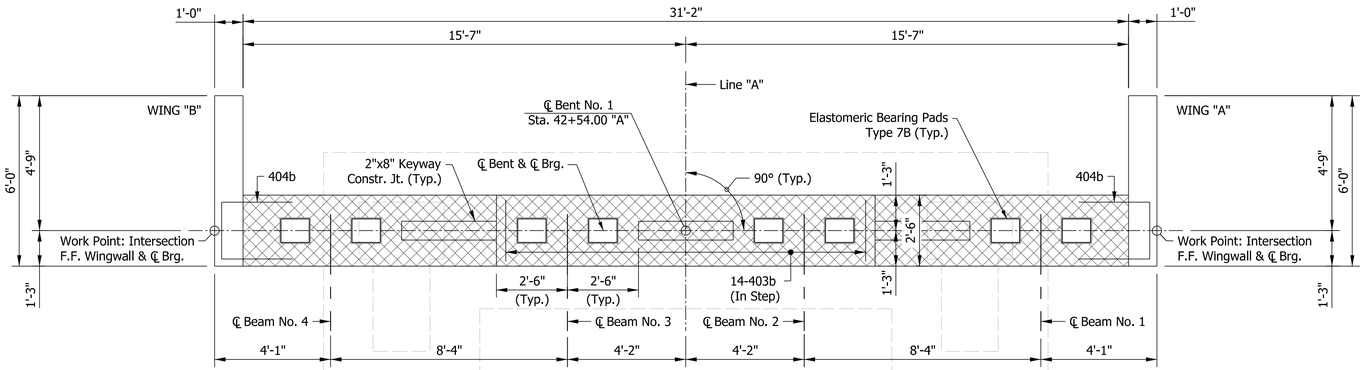
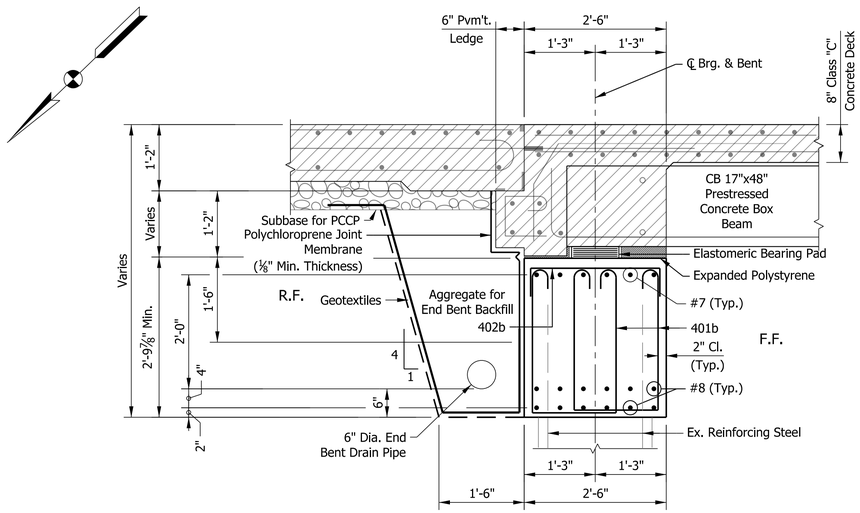


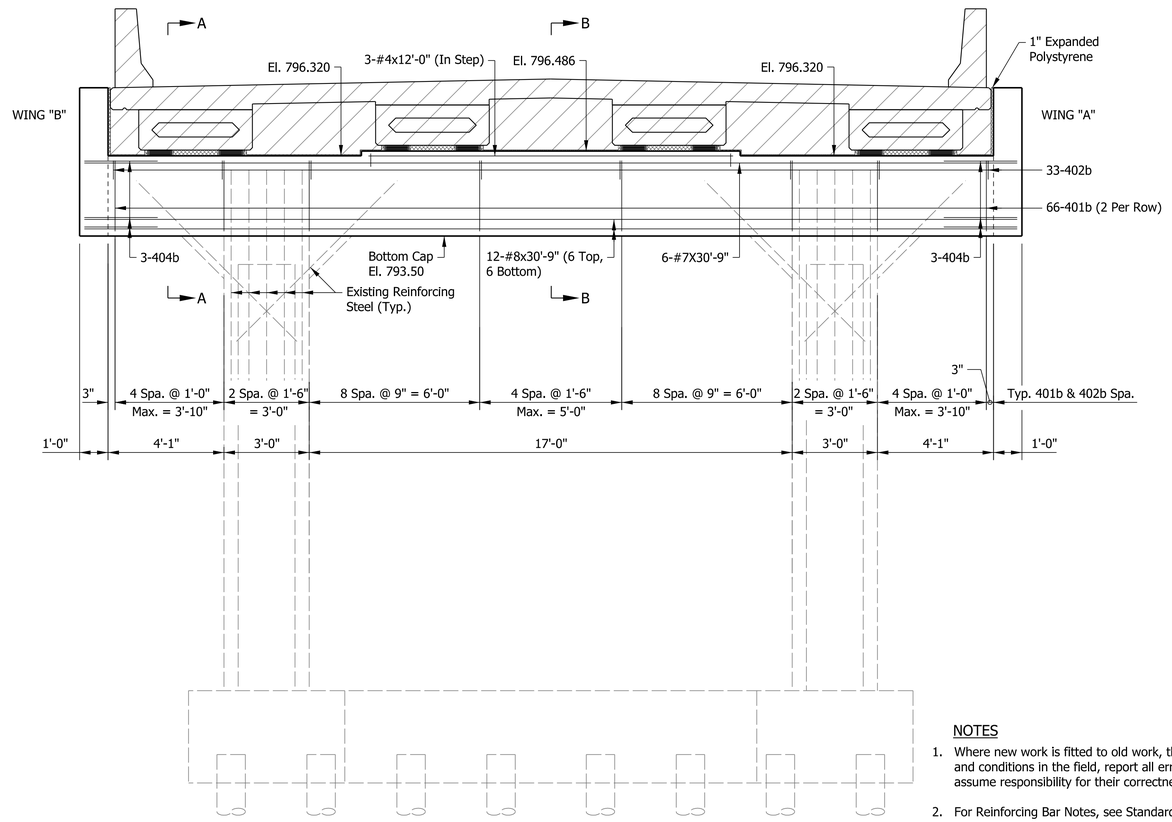
PLAN @ ABUTMENT NO. 1

Scale: ⅜" = 1'-0"



SECTION A-A

Scale: ¾" = 1'-0"



ELEVATION @ ABUTMENT NO. 1

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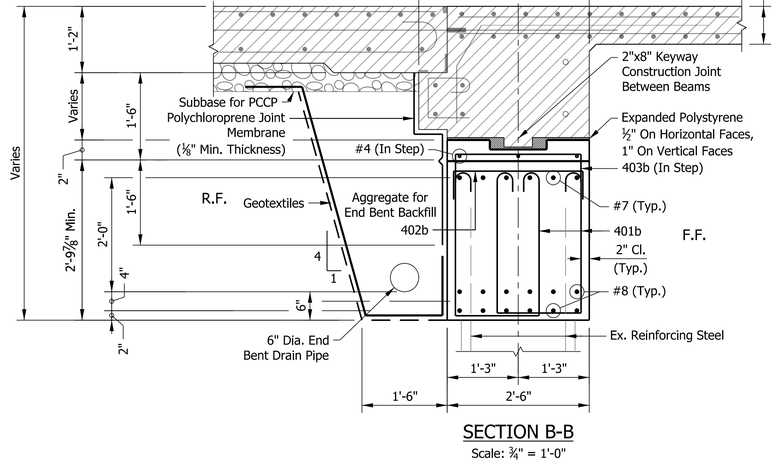

SECTION B-B

Scale: ¾" = 1'-0"

NOTES

1. Where new work is fitted to old work, the Contractor shall check all dimensions and conditions in the field, report all errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.
2. For Reinforcing Bar Notes, see Standard Drawing E703-BRST-01.
3. All reinforcing steel to be epoxy coated.
4. Hatched area to be poured with superstructure.
5. Cross-Hatched areas indicate limits of Expanded Polystyrene cut out to clear elastomeric bearing pads by ½" on all sides. For additional detail and treatment of keyway construction joint, see Section B-B.
6. For Elastomeric Bearing Pad Details, see Std Drawing No. E 726-BEBP-01.
7. Surface Seal all exposed surfaces of wings.
8. For additional details and Bill Of Materials, see Sheet No. 11.
9. For end bent backfill limits and drain pipe details, see Standard Drawing E211-BFIL-04.
10. Epoxy coat existing reinforcing that is incorporated into new concrete.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: KMP DRAWN: TMT

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INDIANA
DEPARTMENT OF TRANSPORTATION

ABUTMENT NO. 1 DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
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CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:34 AM By: thomatm Pen: Transportation.tbl

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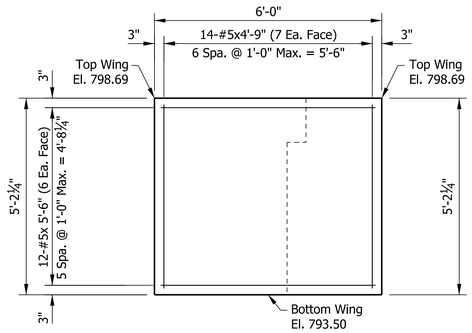
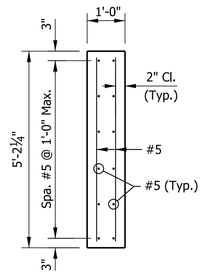


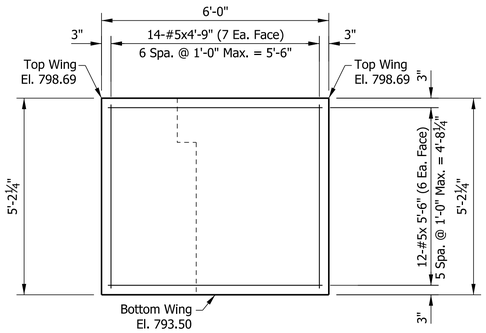
ELEVATION @ WING "B"

Scale: ½" = 1'-0"



TYPICAL WINGWALL SECTION

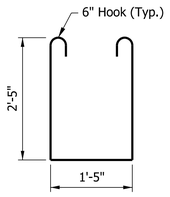
Scale: ½" = 1'-0"



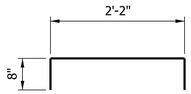
ELEVATION @ WING "A"

Scale: ½" = 1'-0"

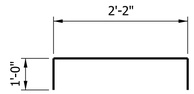
BILL OF MATERIALS ABUTMENT NO. 1			
EPOXY COATED REINFORCING STEEL			
Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
#8	12	30'-9"	
TOTAL #8 BARS			985
#7	6	30'-9"	
TOTAL #7 BARS			377
#5	24	5'-6"	
#5	28	4'-9"	
TOTAL #5 BARS			276
401b	66	7'-3"	
402b	33	3'-6"	
403b	14	4'-2"	
404b	6	7'-0"	
TOTAL #4 BARS			464
TOTAL EPOXY COATED REINF.			2,103
CONCRETE			
CLASS "A" IN SUBSTRUCTURE			10.7 CYS
MISCELLANEOUS			
AGG. FOR END BENT BACKFILL			10 CYS
GEOTEXTILE FOR UNDERDRAIN TYPE 3			30 SYS
PIPE, END BENT DRAIN, 6 IN.			51 LFT



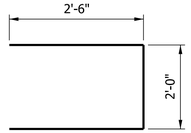
401b x 7'-3"



402b x 3'-6"



403b x 4'-2"



404b x 7'-0"

NOTES

1. Where new work is fitted to old work, the Contractor shall check all dimensions and conditions in the field, report all errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.
2. For Reinforcing Bar Notes, see Std. Dwg. E 703-BRST-01.
3. All reinforcing bars shall be Epoxy Coated.
4. Epoxy coat existing reinforcing that is incorporated into new concrete.
5. Surface Seal all exposed surfaces of wings.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: KMP DRAWN: TMT

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INDIANA
DEPARTMENT OF TRANSPORTATION

ABUTMENT NO. 1 DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
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	SHEETS
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CONTRACT	PROJECT
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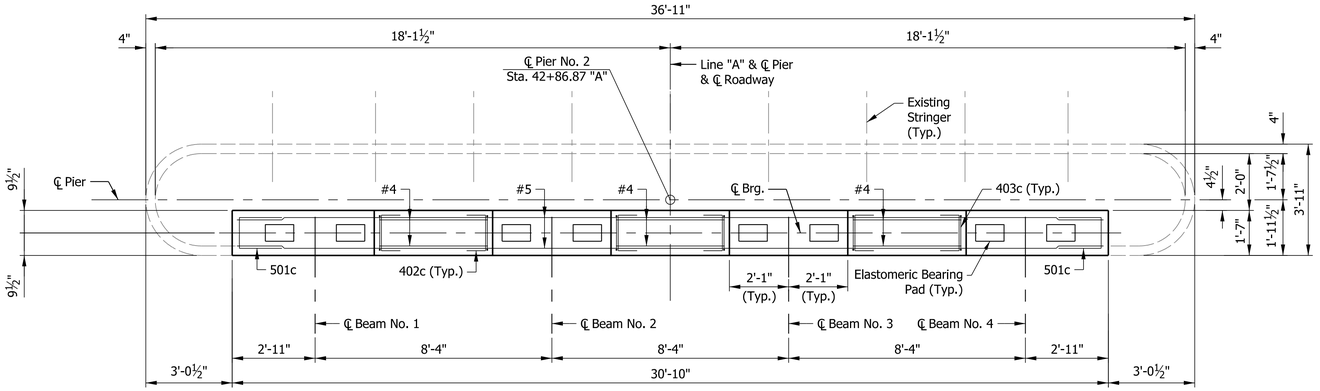
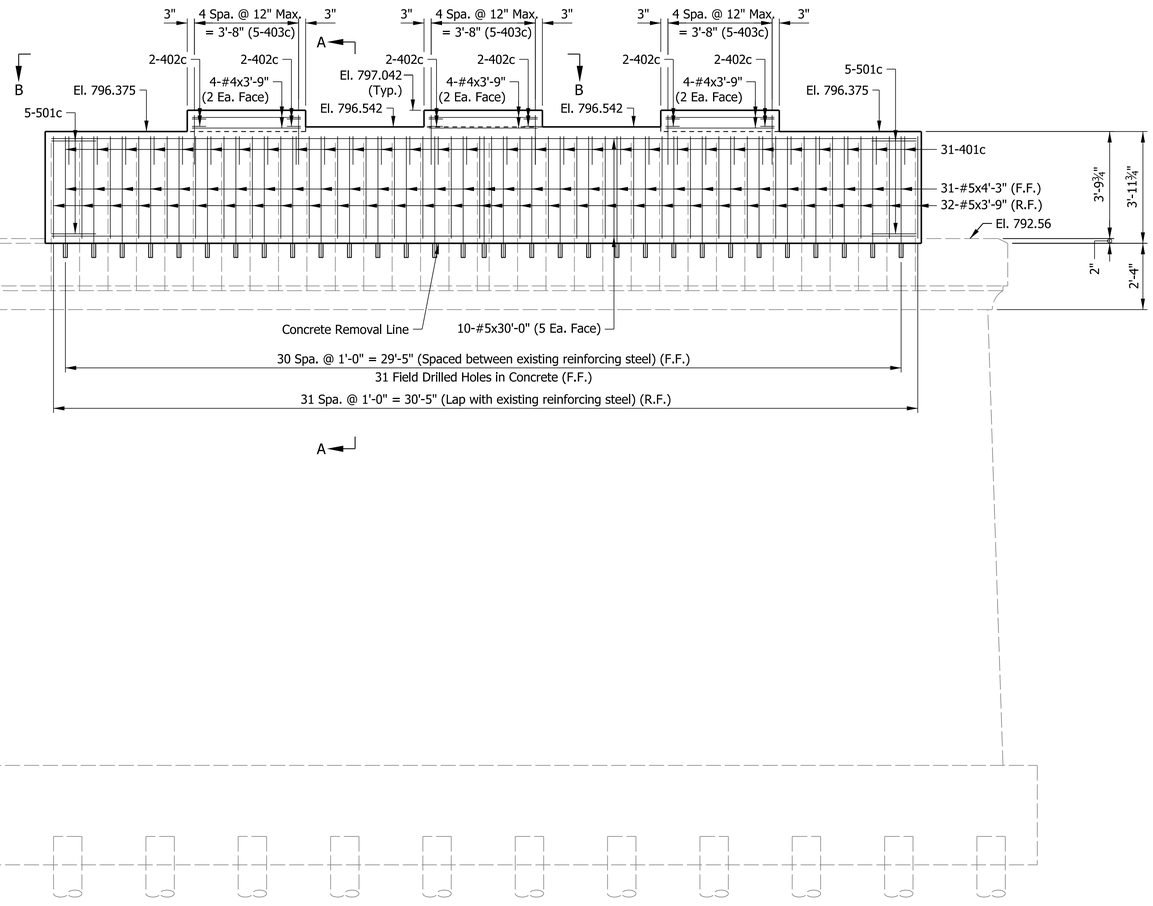
Plot: 2/14/2020 10:31:36 AM By: thomatm Pen: Transportation.tbl



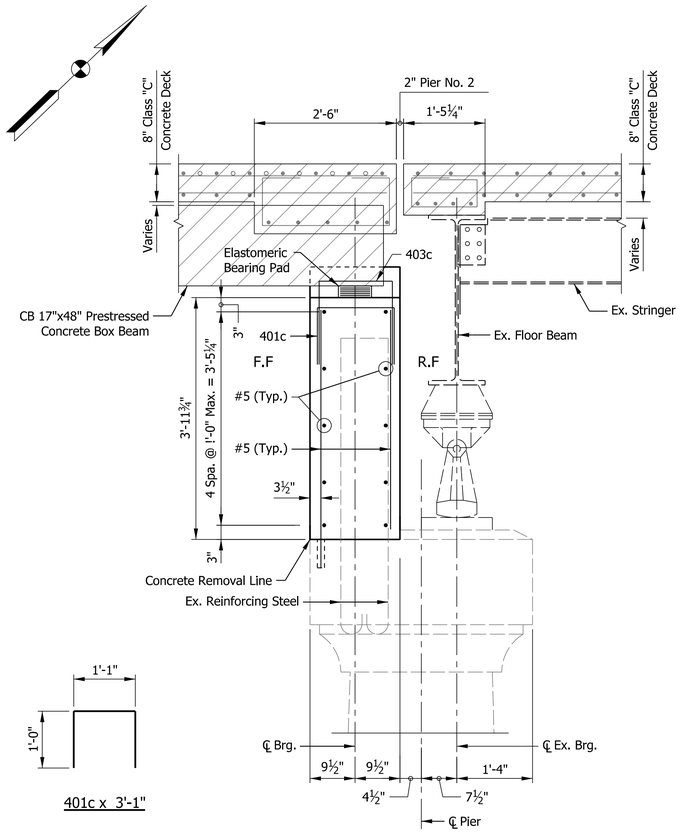
PLAN @ PIER NO. 2

Scale: ⅜" = 1'-0"



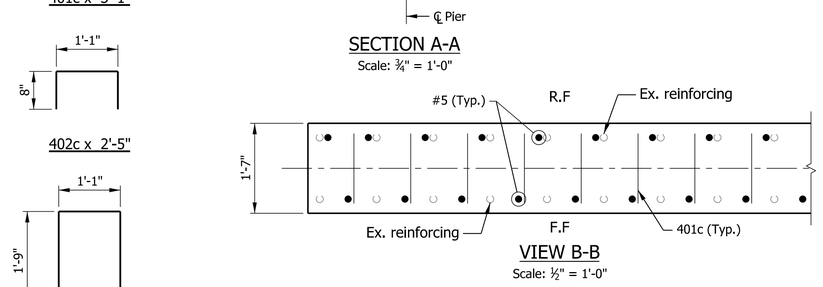
ELEVATION @ PIER NO. 2

Scale: ⅜" = 1'-0"



SECTION A-A

Scale: ¾" = 1'-0"



VIEW B-B

Scale: ½" = 1'-0"

BILL OF MATERIALS
PIER NO. 2

Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
EPOXY COATED REINFORCING STEEL			
501c	10	2'-9"	
#5	10	30'-6"	
#5	31	4'-3"	
#5	32	3'-9"	
TOTAL #5 BARS			609
401c	31	3'-1"	
402c	12	2'-5"	
403c	15	4'-7"	
TOTAL #4 BARS			129
TOTAL EPOXY COATED REINF.			739
CONCRETE			
CLASS "A" IN SUBSTRUCTURE			7.9 CYS
MISCELLANEOUS			
SURFACE SEAL			255 SFT
FIELD DRILLED HOLES IN CONCRETE			31 EACH

NOTES

1. Where new work is fitted to old work, the Contractor shall check all dimensions and conditions in the field, report all errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.
2. Intentionally Roughen Existing Concrete ¼" where it meets new concrete.
3. For Reinforcing Bar Notes, see Standard Drawing E703-BRST-01.
4. All reinforcing steel to be epoxy coated.
5. Field drilled holes in concrete shall extend to a depth required to embed a #5 bar 6" minimum with an approved anchor system having a minimum pullout equal to 18,600 Lbs. Existing Pier Cap Reinforcing shall not be cut during installation of anchors.
6. For Elastomeric Bearing Pad Details, see sheet 21.
7. Surface Seal the top and vertical surfaces of pier cap.
7. Install embedded galvanize anodes (EGA) on alternating reinforcing bars at interface of new concrete to old in new beam seat construction. Install EGAs on exposed reinforcing within locations of pier patches as directed by the engineer.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: NRT DRAWN: TMT

CHECKED: KMP CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

PIER NO. 2 DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	12 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:41 AM By: thomatm Pen: Transportation.tbl

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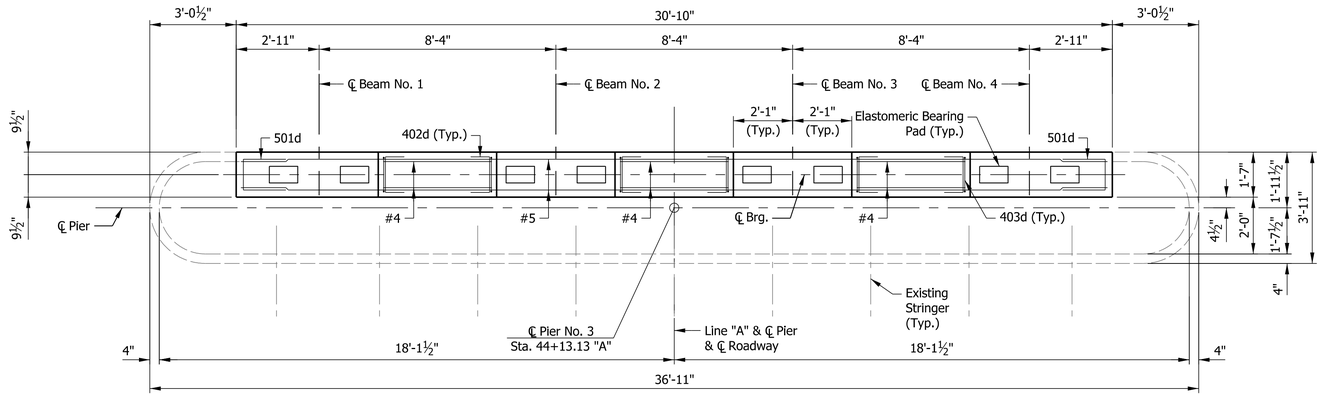
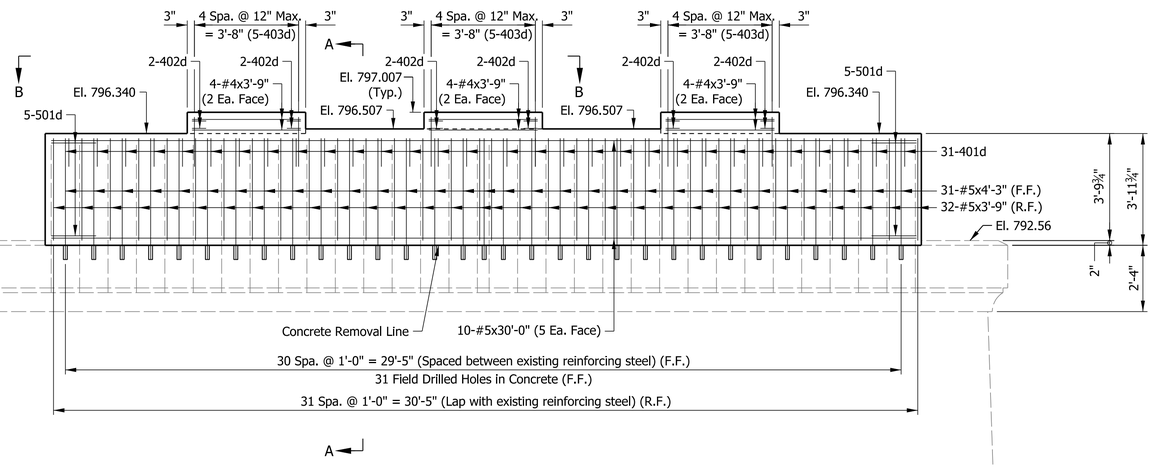


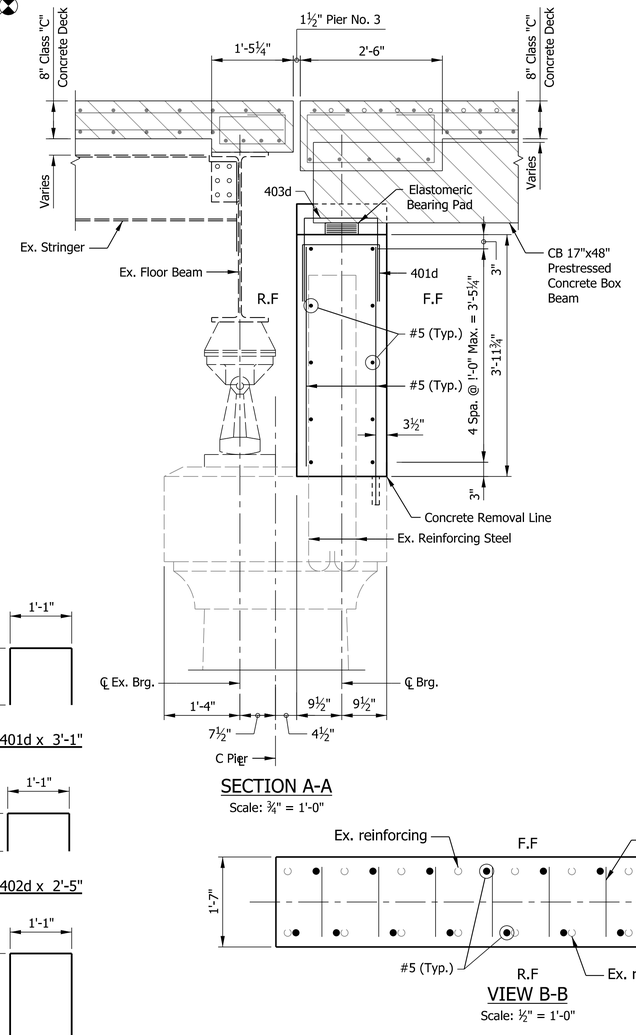
PLAN @ PIER NO. 3

Scale: ⅜" = 1'-0"



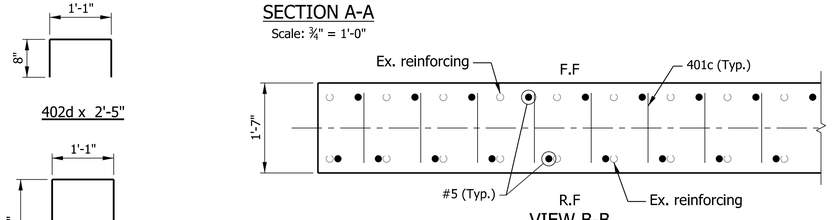
ELEVATION @ PIER NO. 3

Scale: ⅜" = 1'-0"



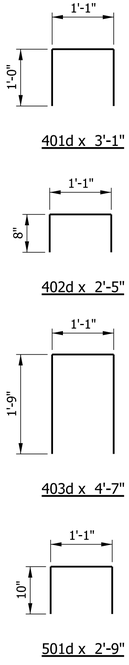
SECTION A-A

Scale: ¾" = 1'-0"



VIEW B-B

Scale: ½" = 1'-0"



BILL OF MATERIALS PIER NO. 3

EPOXY COATED REINFORCING STEEL			
Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
501d	10	2'-9"	
#5	10	30'-6"	
#5	31	4'-3"	
#5	32	3'-9"	
TOTAL #5 BARS			609
401d	31	3'-1"	
402d	12	2'-5"	
403d	15	4'-7"	
TOTAL #4 BARS			129
TOTAL EPOXY COATED REINF.			739
CONCRETE			
CLASS "A" IN SUBSTRUCTURE			7.8 CYS
MISCELLANEOUS			
SURFACE SEAL			255 SFT
FIELD DRILLED HOLES IN CONCRETE			31 EACH

NOTES

1. Where new work is fitted to old work, the Contractor shall check all dimensions and conditions in the field, report all errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.
2. Intentionally Roughen Existing Concrete ¼" where it meets new concrete.
3. For Reinforcing Bar Notes, see Standard Drawing E703-BRST-01.
4. All reinforcing steel to be epoxy coated.
5. Field drilled holes in concrete shall extend to a depth required to embed a #5 bar 6" minimum with an approved anchor system having a minimum pullout equal to 18,600 Lbs. Existing Pier Cap Reinforcing shall not be cut during installation of anchors.
6. For Elastomeric Bearing Pad Details, see sheet 21.
7. Surface Seal the top and vertical surfaces of pier cap.
7. Install embedded galvanize anodes (EGA) on alternating reinforcing bars at interface of new concrete to old in new beam seat construction. Install EGAs on exposed reinforcing within locations of pier patches as directed by the engineer.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: NRT DRAWN: TMT

CHECKED: KMP CHECKED: KMP

INDIANA DEPARTMENT OF TRANSPORTATION

PIER NO. 3 DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	13 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:46 AM By: thomatm Pen: Transportation.tbl

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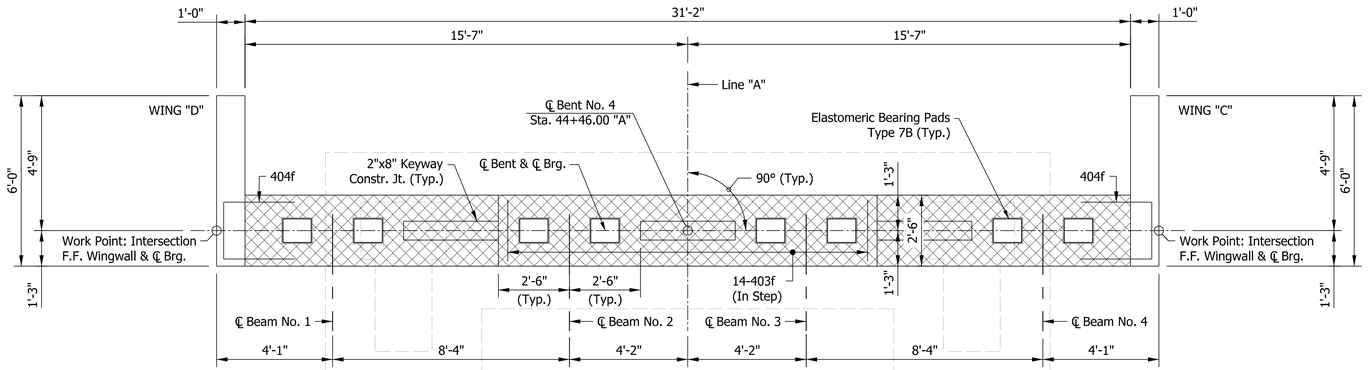
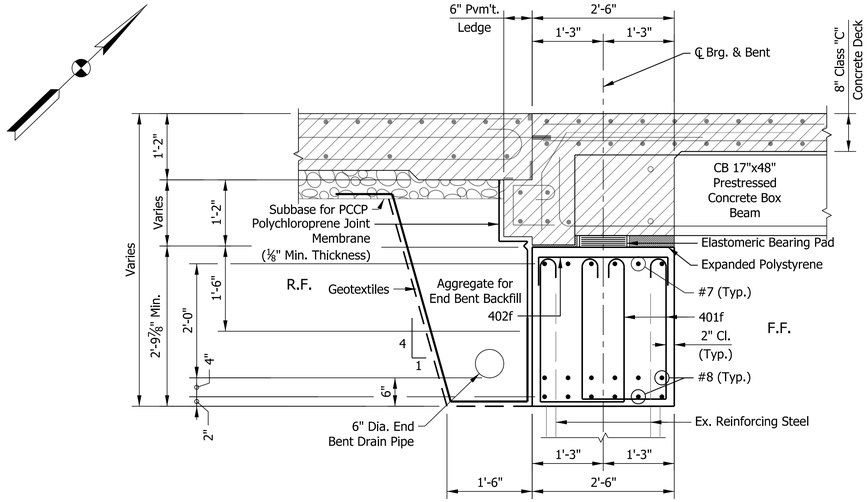


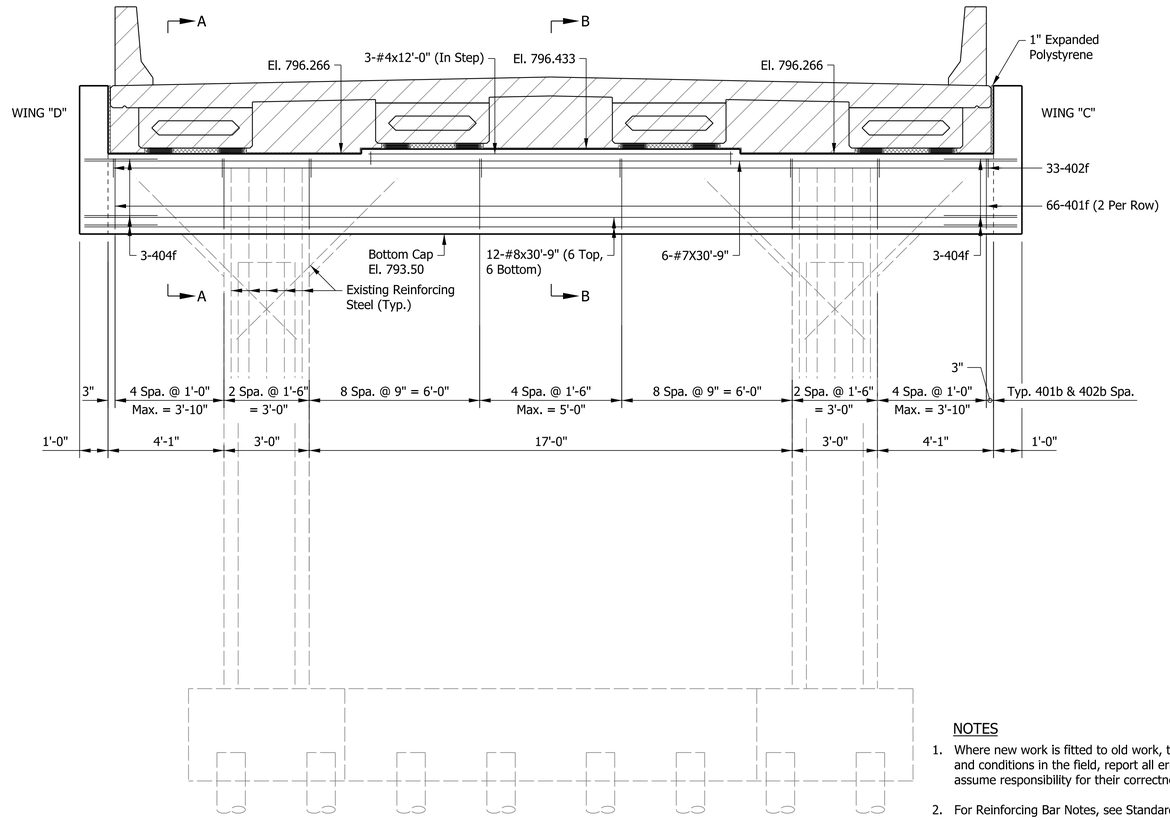
PLAN @ ABUTMENT NO. 4

Scale: ⅜" = 1'-0"



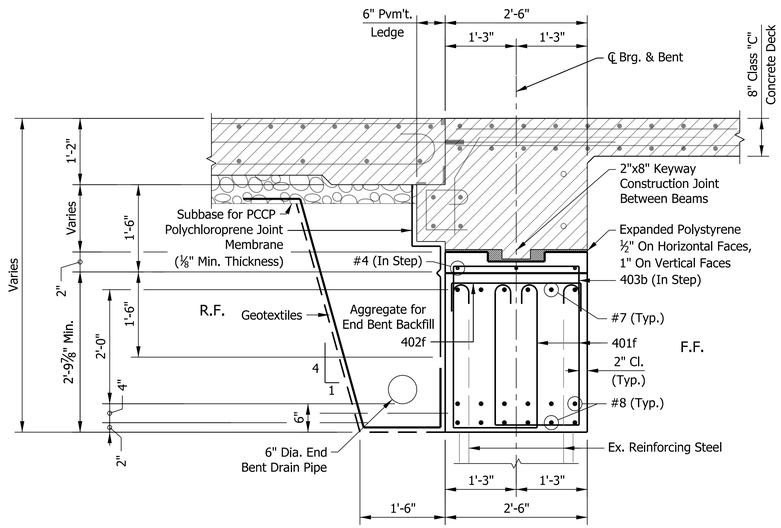
SECTION A-A

Scale: ¾" = 1'-0"



ELEVATION @ ABUTMENT NO. 4

Scale: ⅜" = 1'-0"



SECTION B-B

Scale: ¾" = 1'-0"

NOTES

1. Where new work is fitted to old work, the Contractor shall check all dimensions and conditions in the field, report all errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.
2. For Reinforcing Bar Notes, see Standard Drawing E703-BRST-01.
3. All reinforcing steel to be epoxy coated.
4. Hatched area to be poured with superstructure.
5. Cross-Hatched areas indicate limits of Expanded Polystyrene cut out to clear elastomeric bearing pads by ½" on all sides. For additional detail and treatment of keyway construction joint, see Section B-B.
6. For Elastomeric Bearing Pad Details, see Std Drawing No. E 726-BEBP-01.
7. Surface Seal all exposed surfaces of wings.
8. For additional details and Bill Of Materials, see Sheet No. 15.
9. For end bent backfill limits and drain pipe details, see Standard Drawing E211-BFIL-04.
10. Epoxy coat existing reinforcing that is incorporated into new concrete.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: KMP DRAWN: TMT

CHECKED: TDJ CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

ABUTMENT NO. 4 DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	14 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:51 AM By: thomatm Pen: Transportation.tbl

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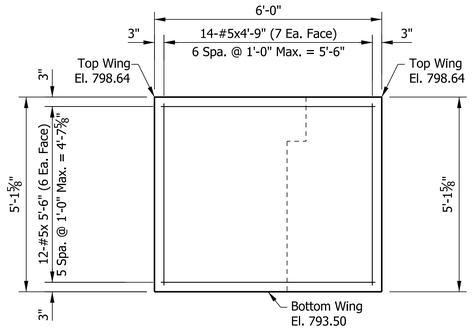
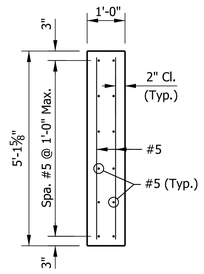
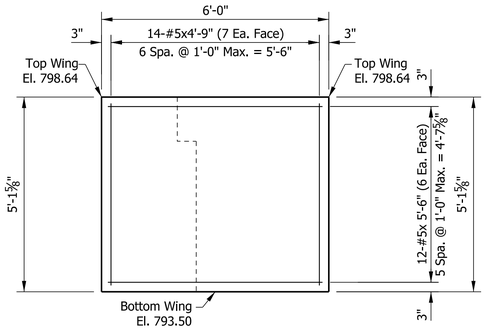


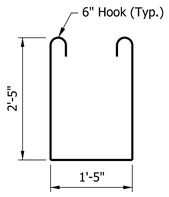
ELEVATION @ WING "D"
Scale: ½" = 1'-0"



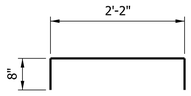
TYPICAL WINGWALL SECTION
Scale: ½" = 1'-0"



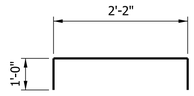
ELEVATION @ WING "C"
Scale: ½" = 1'-0"



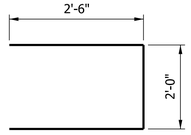
401f x 7'-3"



402f x 3'-6"



403f x 4'-2"



404f x 7'-0"

BILL OF MATERIALS ABUTMENT NO. 4			
EPOXY COATED REINFORCING STEEL			
Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
#8	12	30'-9"	
TOTAL #8 BARS			985
#7	6	30'-9"	
TOTAL #7 BARS			377
#5	24	5'-6"	
#5	28	4'-9"	
TOTAL #5 BARS			276
401f	66	7'-3"	
402f	33	3'-6"	
403f	14	4'-2"	
404f	6	7'-0"	
TOTAL #4 BARS			464
TOTAL EPOXY COATED REINF.			2,103
CONCRETE			
CLASS "A" IN SUBSTRUCTURE			10.5 CYS
MISCELLANEOUS			
AGG. FOR END BENT BACKFILL			10 CYS
GEOTEXTILE FOR UNDERDRAIN TYPE 3			30 SYS
PIPE, END BENT DRAIN, 6 IN.			51 LFT

NOTES

1. Where new work is fitted to old work, the Contractor shall check all dimensions and conditions in the field, report all errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.
2. For Reinforcing Bar Notes, see Std. Dwg. E 703-BRST-01.
3. All reinforcing bars shall be Epoxy Coated.
4. Epoxy coat existing reinforcing that is incorporated into new concrete.
5. Surface Seal all exposed surfaces of wings.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: KMP DRAWN: TMT

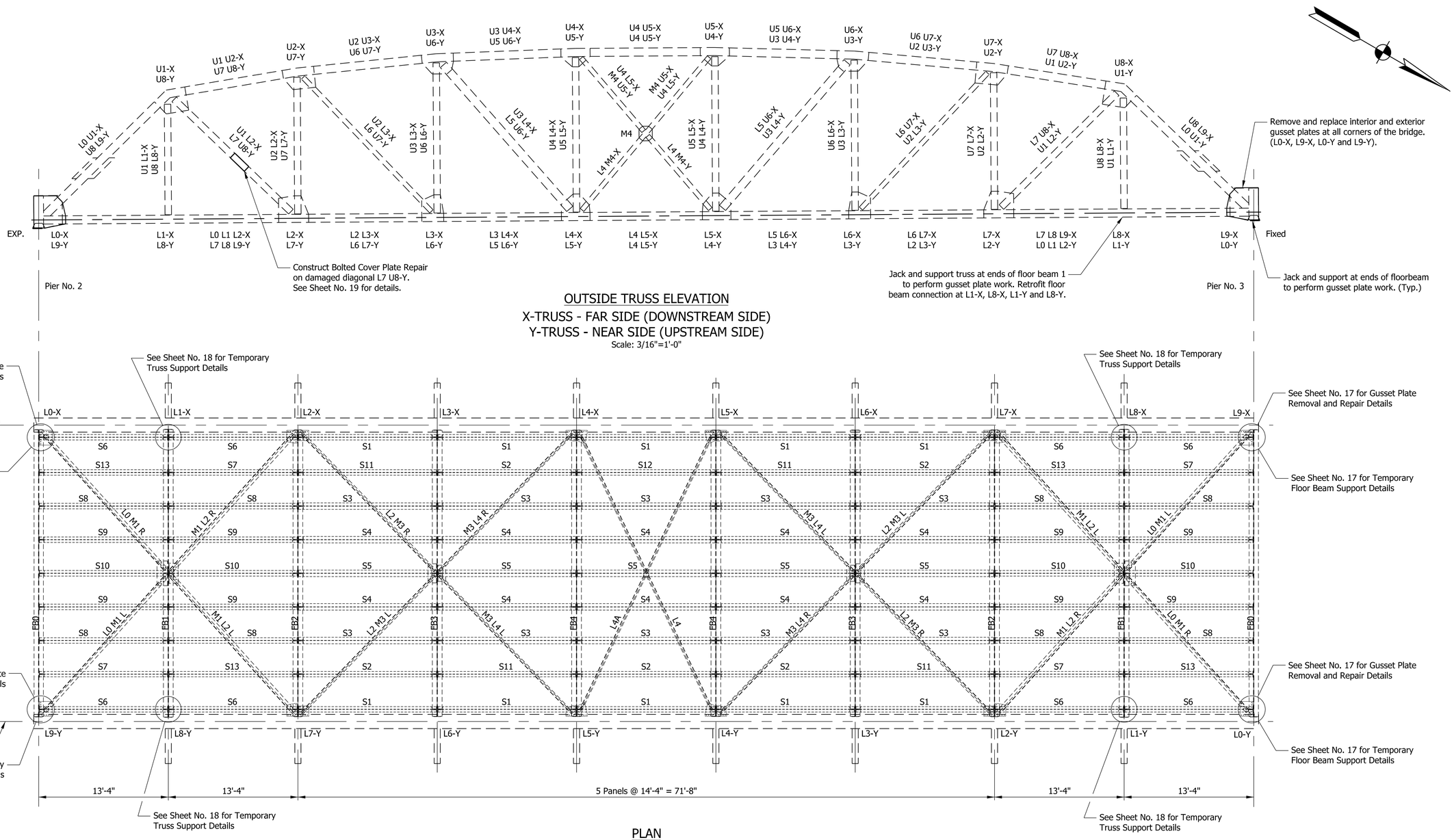
CHECKED: TDJ CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

ABUTMENT NO. 4 DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	15 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:53 AM By: thomatm Pen: Transportation.tbl



OUTSIDE TRUSS ELEVATION

X-TRUSS - FAR SIDE (DOWNSTREAM SIDE)
Y-TRUSS - NEAR SIDE (UPSTREAM SIDE)
Scale: 3/16"=1'-0"

PLAN

Scale: 3/16"=1'-0"

NOTES

1. For additional Truss details see archived std. superstructure details, drawing std. No. 1532 dated May 26, 1939 and revised Nov. 12, 1940.
2. Replace deteriorated members or portions of members as directed by the Engineer following structural inspection during construction. See Special Provisions.

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INDIANA DEPARTMENT OF TRANSPORTATION

STEEL TRUSS DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
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CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:31:58 AM By: thomatm Pen: Transportation.tbl

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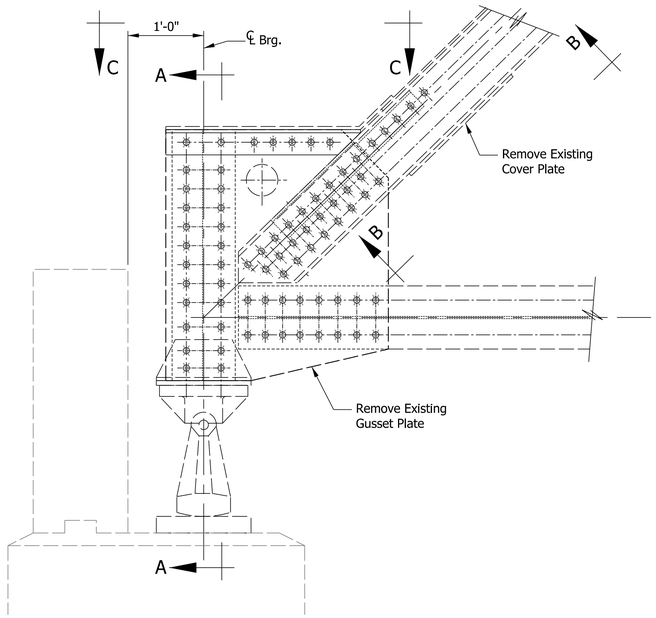
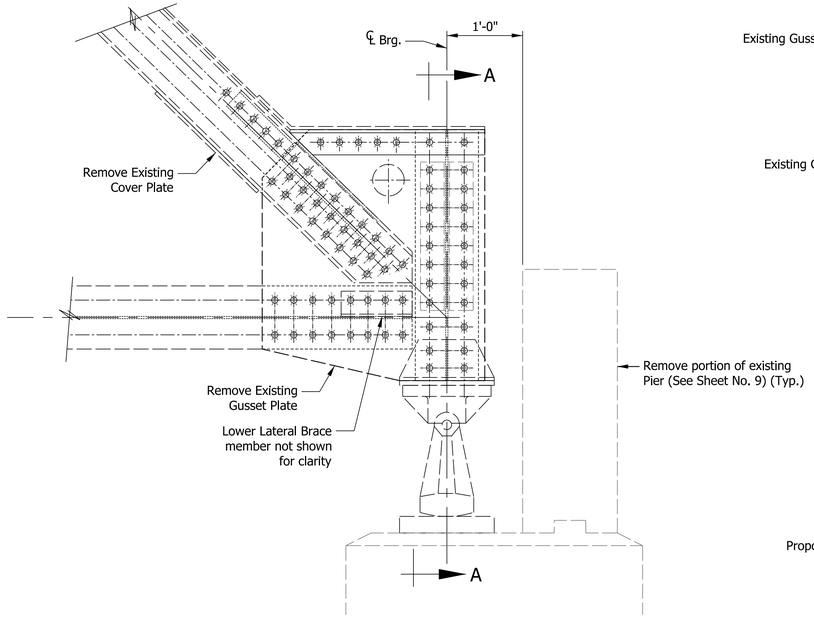


EXISTING OUTSIDE GUSSET PLATE ELEVATION

(L9-Y SHOWN, L9-X SAME
L0-Y AND L0-X SIMILAR BY OPPOSITE HAND)

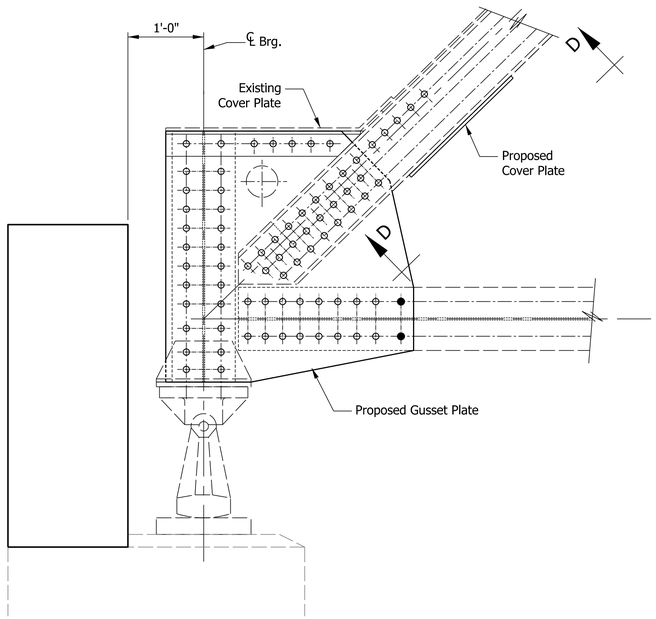
Scale: 1"=1'-0"



EXISTING INSIDE GUSSET PLATE ELEVATION

(L9-Y SHOWN, L9-X SAME
L0-Y AND L0-X SIMILAR BY OPPOSITE HAND)

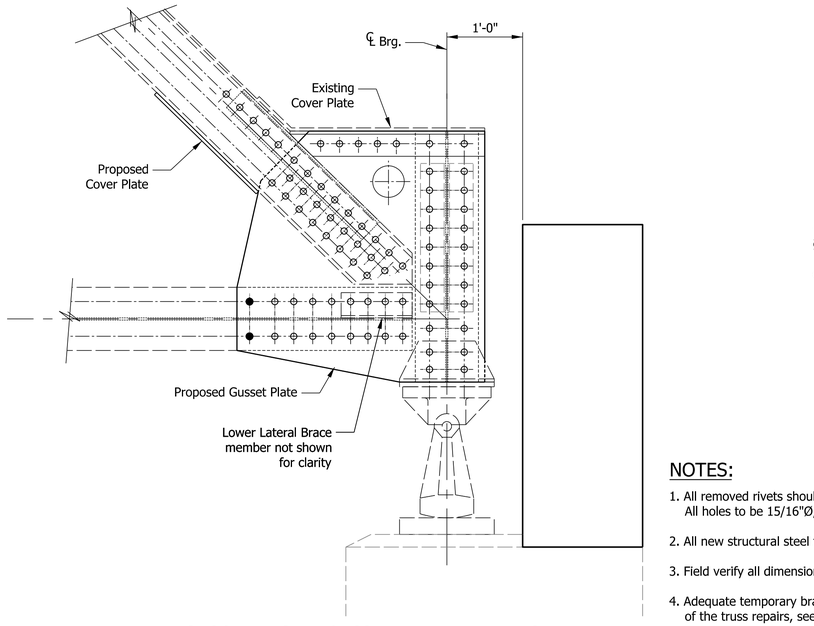
Scale: 1"=1'-0"



PROPOSED OUTSIDE GUSSET PLATE ELEVATION

(L9-Y SHOWN, L9-X SAME
L0-Y AND L0-X SIMILAR BY OPPOSITE HAND)

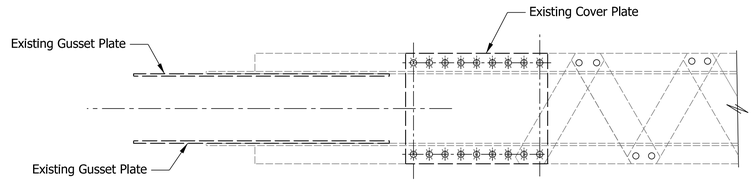
Scale: 1"=1'-0"



PROPOSED INSIDE GUSSET PLATE ELEVATION

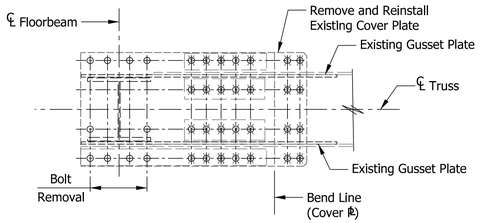
(L9-Y SHOWN, L9-X SAME
L0-Y AND L0-X SIMILAR BY OPPOSITE HAND)

Scale: 1"=1'-0"



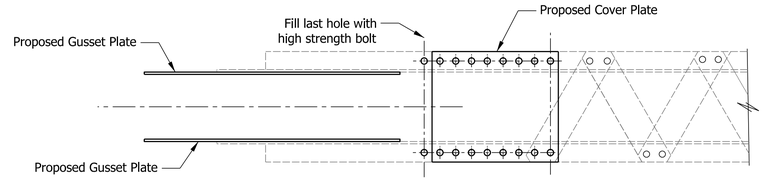
SECTION "B-B"

Scale: 1"=1'-0"



SECTION "C-C"

Scale: 1"=1'-0"



SECTION "D-D"

Scale: 1"=1'-0"

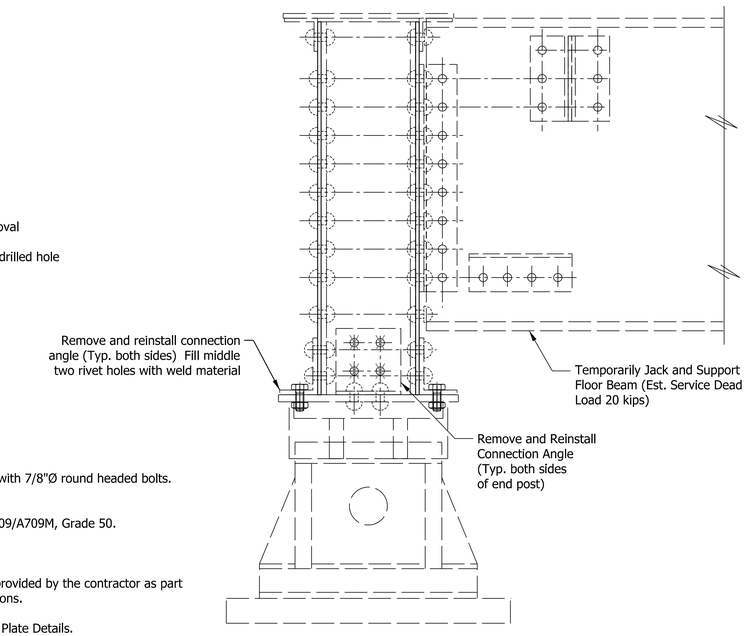
LEGEND

⊗ - Rivet removal

● - New field drilled hole

NOTES:

1. All removed rivets should be replaced with 7/8"Ø round headed bolts. All holes to be 15/16"Ø, unless noted.
2. All new structural steel to be ASTM A709/A709M, Grade 50.
3. Field verify all dimensions.
4. Adequate temporary bracing shall be provided by the contractor as part of the truss repairs, see special provisions.
5. See Sheet No. 18 for Proposed Gusset Plate Details.
6. Members removed but not replaced shall be properly marked and reincorporated into the final connection.



SECTION A-A

(TYPICAL AT EACH END OF FB0)

Scale: 1 1/2"=1'-0"

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RECOMMENDED FOR APPROVAL _____ - DESIGN ENGINEER DATE

DESIGNED: JAS | DRAWN: TMT

CHECKED: KMP | CHECKED: KMP

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DEPARTMENT OF TRANSPORTATION

GUSSET PLATE REPLACEMENT DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	17 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:32:01 AM By: thomatm Pen: Transportation.tbl

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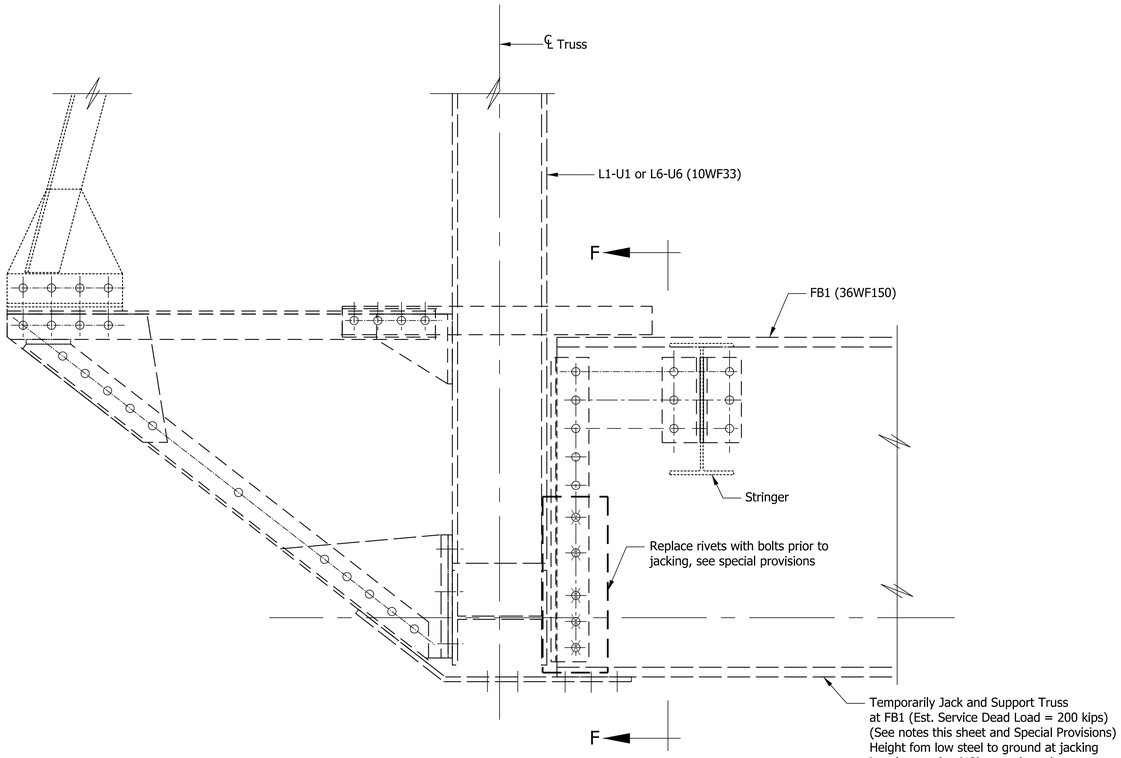
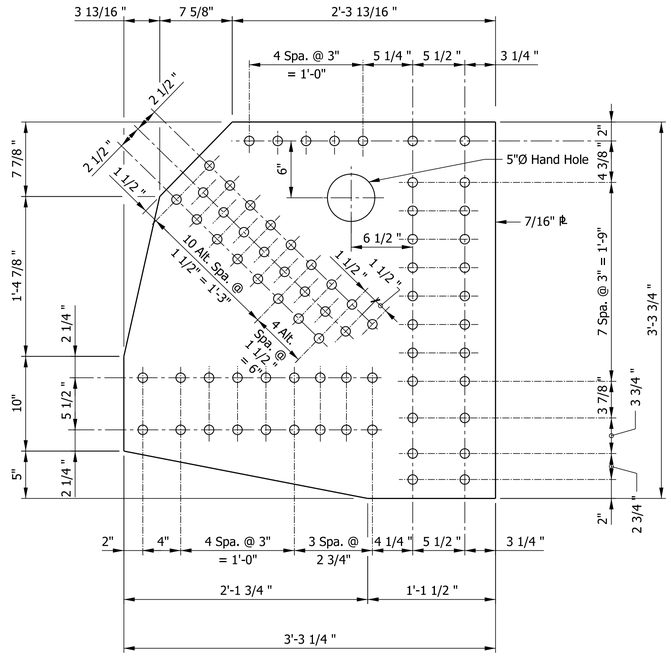


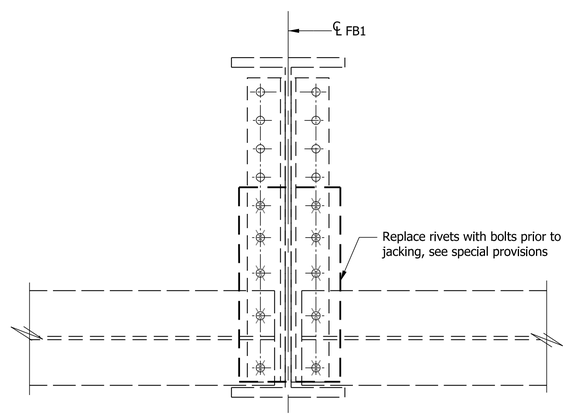
TEMPORARY TRUSS SUPPORT RETROFIT DETAILS

Scale: 1 1/2"=1'-0"



PROPOSED GUSSET PLATE DETAIL

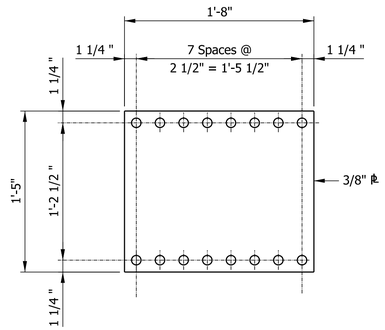
Scale: 1"=1'-0"



SECTION "F-F"

Scale: 1 1/2"=1'-0"

(Stringers Not Shown for Clarity)



PROPOSED COVER PLATE

Scale: 1 1/2"=1'-0"

LEGEND

⊗ - Rivet removal

● - New field drilled hole

NOTES

1. Estimated Dead Load is at each end of FB1 and assumes a sequence of work such that temporary support and gusset replacement occurs only on one side of the creek at a time, with the deck slab completely removed prior to jacking.
2. Alternate jacking and supporting locations and methods may be used if approved by the engineer. (See Special Provisions).
3. All removed rivets should be replaced with 7/8"Ø round headed bolts. All holes to be 15/16"Ø.
4. All new structural steel to be ASTM A709/A709M, Grade 50.
5. Field verify all dimensions.
6. Adequate temporary bracing shall be provided by the contractor as part of the truss repairs, see special provisions.
7. Members removed but not replaced shall be properly marked and reincorporated into the final connection.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: JAS | DRAWN: TMT

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INDIANA DEPARTMENT OF TRANSPORTATION

TEMPORARY SUPPORT DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
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CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:32:04 AM By: thomatm Pen: Transportation.tbl

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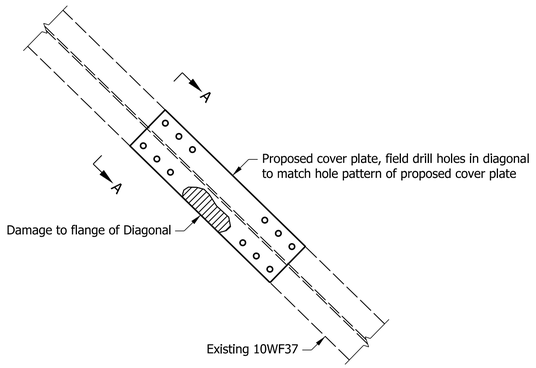
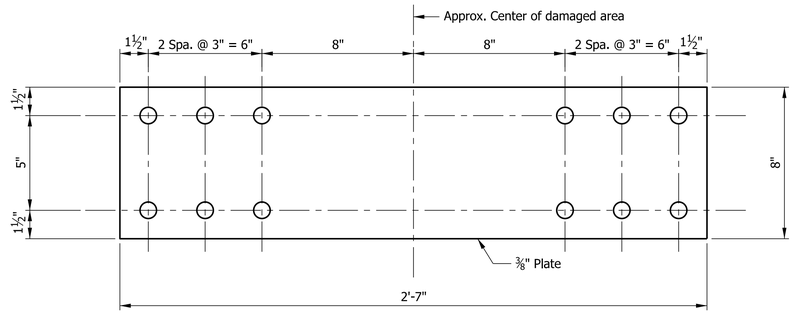
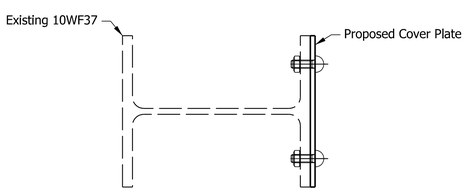


BOLTED COVER PLATE REPAIR (L7 U8-Y)
Scale: 1½" = 1'-0"



PROPOSED COVER PLATE DETAIL
Scale: 3" = 1'-0"



SECTION A-A
Scale: 3" = 1'-0"

NOTES

1. All bolts shall be ⅞"Ø round headed bolts.
   All holes to be $^{15}/_{16}$"Ø, unless noted.
2. All new structural steel to be ASTM A709/A709M, Grade 50.
3. Field verify all dimensions.
4. Repair or replacement of additional members only required if determined by engineer following a structural inspection during construction. See unique special provisions.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: JAS | DRAWN: TMT

CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

COVER PLATE DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	19 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:32:06 AM By: thomatm Pen: Transportation.tbl

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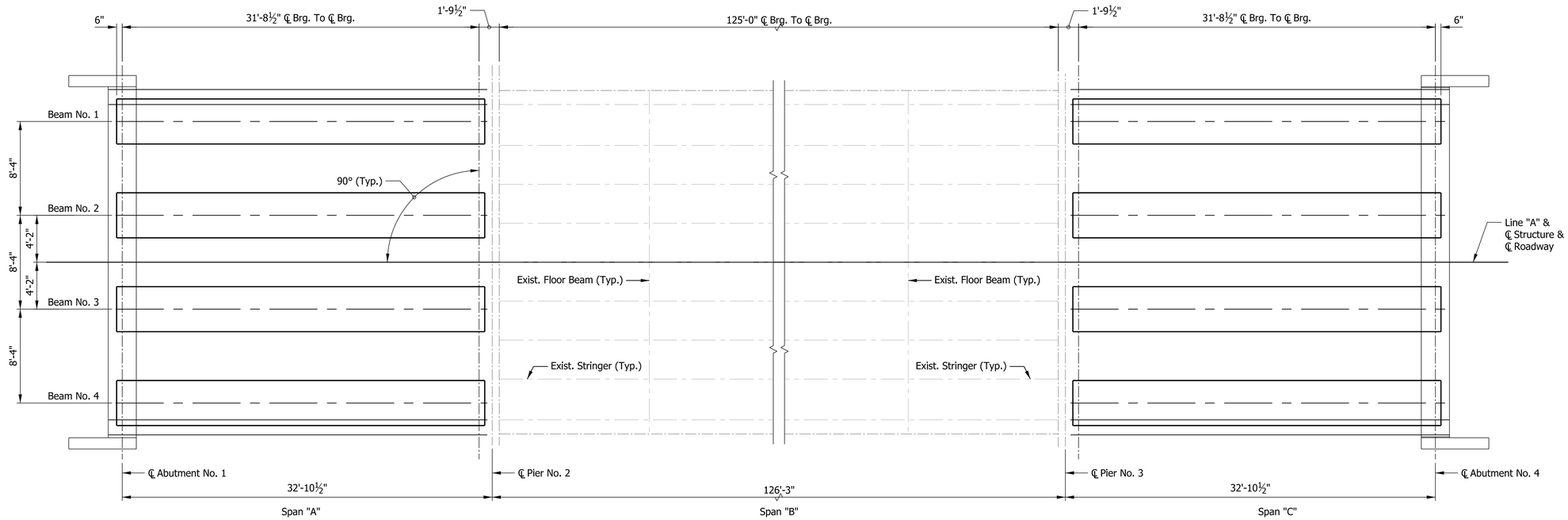
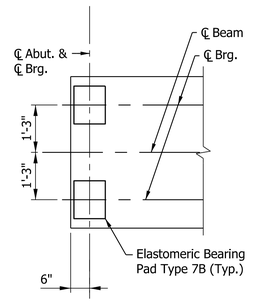


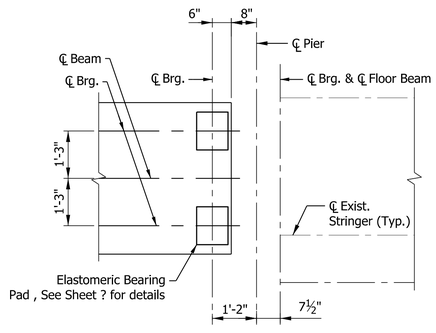
FRAMING PLAN

Scale: 3⁄16" = 1'-0"



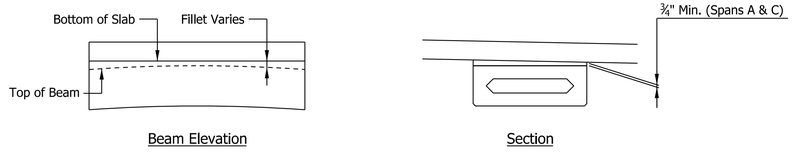
TYPICAL ELASTOMERIC BEARING PAD PLACEMENT AT ABUTMENTS NO. 1 & 4

Scale: ½" = 1'-0"



TYPICAL ELASTOMERIC BEARING PAD PLACEMENT AT PIERS NO. 2 & 3

Scale: ½" = 1'-0"



Beam Elevation

Section

BEAM FILLET DETAIL

No Scale

Residual Camber Table	
	Span "A" & "C"
Initial Beam Camber	¾"
Superimposed Dead Load Deflection	-¼"
Residual Camber	½"

Initial Beam Camber Equals Upward Deflection From Prestressing Force Minus Downward Deflection From Dead Load of the Beam In Inches.

BEAM DESIGN DATA

1. Prestressing steel shall be 0.5" diameter uncoated, special low relaxation, seven-wire strand, 270 ksi, with strand area = 0.167 sq.in.
2. Initial pull per prestressing strand to be 33.82 kips.
3. Concrete strength at release, f'ci = 5,000 psi.
4. Concrete strength at 28 days, f'c = 6,000 psi.
5. Mild reinforcing steel shall be Grade 60 ksi minimum yield strength.

GENERAL BEAM NOTES

1. Beams shall be cast a minimum 30 days prior to pouring the deck.
2. Beams are to be lifted and supported at the bearing points during handling, storage, and transportation. Adequate bracing must be provided at all times during storage, transportation and lifting to resist lateral loads.
3. Allowance should be made in beam length for elastic shortening and grade.
4. For Fabrication Tolerances of Prestressed Beams, see Standard Drawings E 707-BPBF-01 and -04.
5. Beams shall be maintained vertically at all times. Suitable restraint shall be provided to prevent the rotation of the beams, particularly the outside beam, from construction load, such as the weight of the concrete deck, finishing machine, forms, etc.
6. Top of beams shall be scored transversely at about 3" on center with pointed tool. Maximum depth of scoring should be ¼".
7. The ends of the beams at the End Bents shall be cast so that the end of the beam is vertical when placed in final grade condition.
8. Acute angles of box beams shall be chamfered 3" (by Manufacturer).
9. Sealer on the outside face of exterior beams to be done by the fabricator in the shop. Do not rub.

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FRAMING PLAN

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
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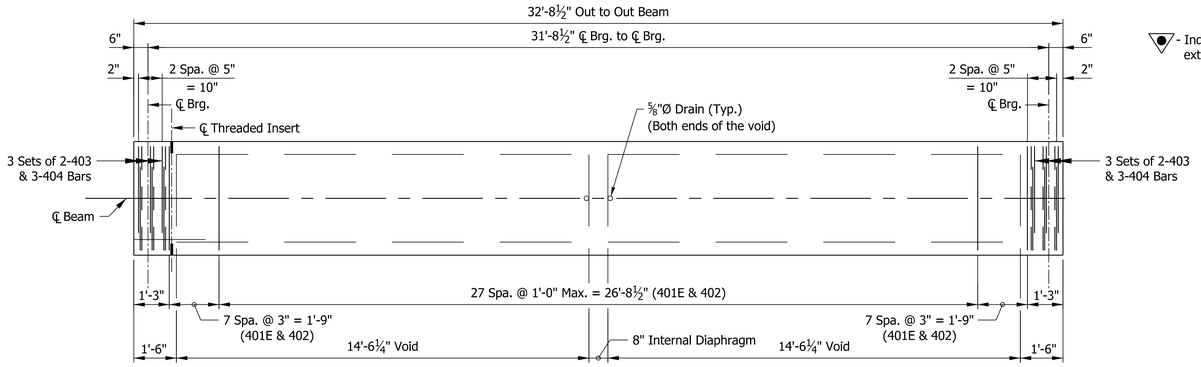
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:32:08 AM By: thomatm Pen: Transportation.tbl

Z:\Infra\2016\D160355.12 - US 421 over S Fork Wildc\CAD\Production Drawings\D160355.12_BR_Framing Plans.01.dgn

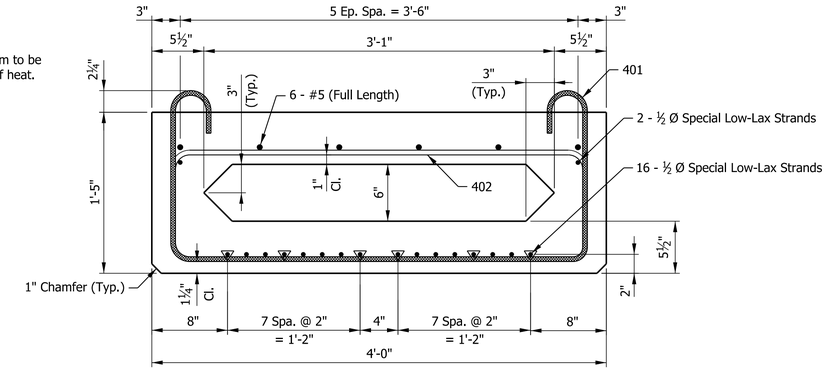


Abutment No. 1 or 4 — Pier No. 2 or 3

BEAM PLAN - SPANS "A" & "C"

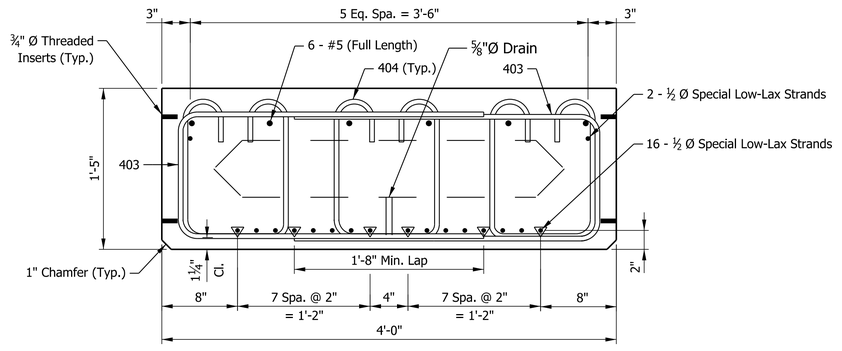
(Span "A" Shown, Span "C" same by 180° Rotation)
No Scale

▽ - Indicates Strands at each end of beam to be extended and bent without the use of heat.



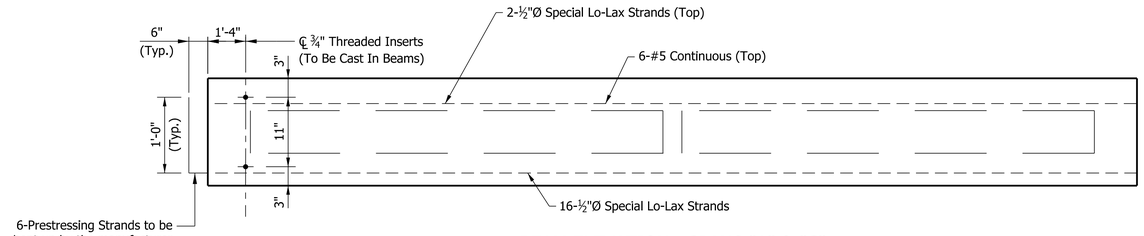
TYPICAL SECTION THRU BOX BEAM

Scale: 1½" = 1'-0"



TYPICAL SECTION THRU END OF BOX BEAM

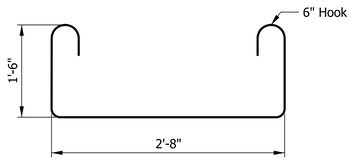
Scale: 1½" = 1'-0"



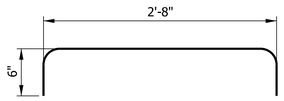
Abutment No. 1 or 4 — Pier No. 2 or 3

BEAM ELEVATION - SPANS "A" & "C"

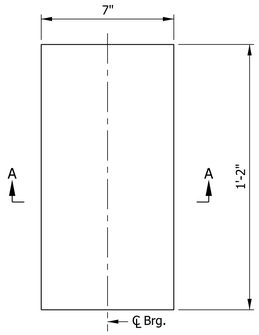
(Span "A" Shown, Span "C" same by 180° Rotation)
No Scale



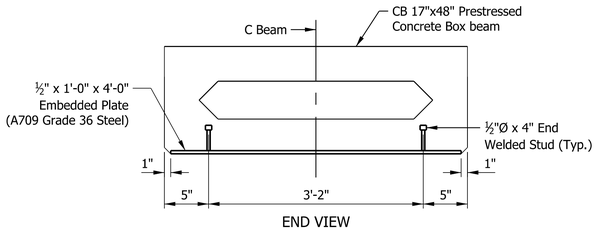
401E x 6'-8"



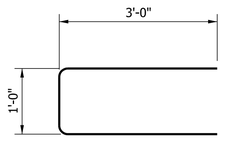
402 x 3'-8"



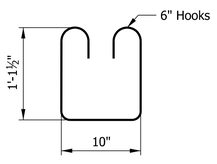
ELASTOMERIC BEARING PAD PLAN



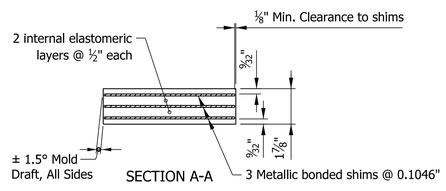
END VIEW



403 x 7'-0"

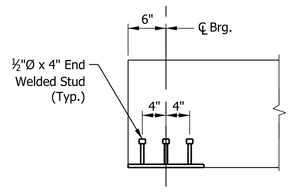


404 x 4'-1"



ELASTOMERIC BEARING PAD DETAILS

Scale: 3" = 1'-0"



SIDE VIEW

EMBEDDED BEARING PLATE DETAILS

Scale: 1" = 1'-0"

NOTES

1. For General Beam Notes, See Sheet 20.
2. For Framing Plan, See Sheet 20.
3. For Beam Design Data, See Sheet 20.
4. For Reinforcing bar notes, See Standard Drawing E703-BRST-01.
5. Reinforcing bars designated (E) shall be epoxy coated.
6. All mild Reinforcing Steel to be Grade 60.
7. Design Method: AASHTO Method A. Bearing Designed per AASHTO method A with Durometer Hardness 55 (±5).
8. Fabricate the Elastomeric Bearing Pads to the Design and Dimensions shown on the plans.
9. Finish Bearing Surface shall be clean and free of loose material before placing Bearing Pads.
10. Plates and Embedded Bearing Plates shall be Hot Dipped Galvanized in accordance with requirements of A.S.T.M. A123 or A153.

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BEAM DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	21 of 30
CONTRACT	PROJECT
B-42017	1593276

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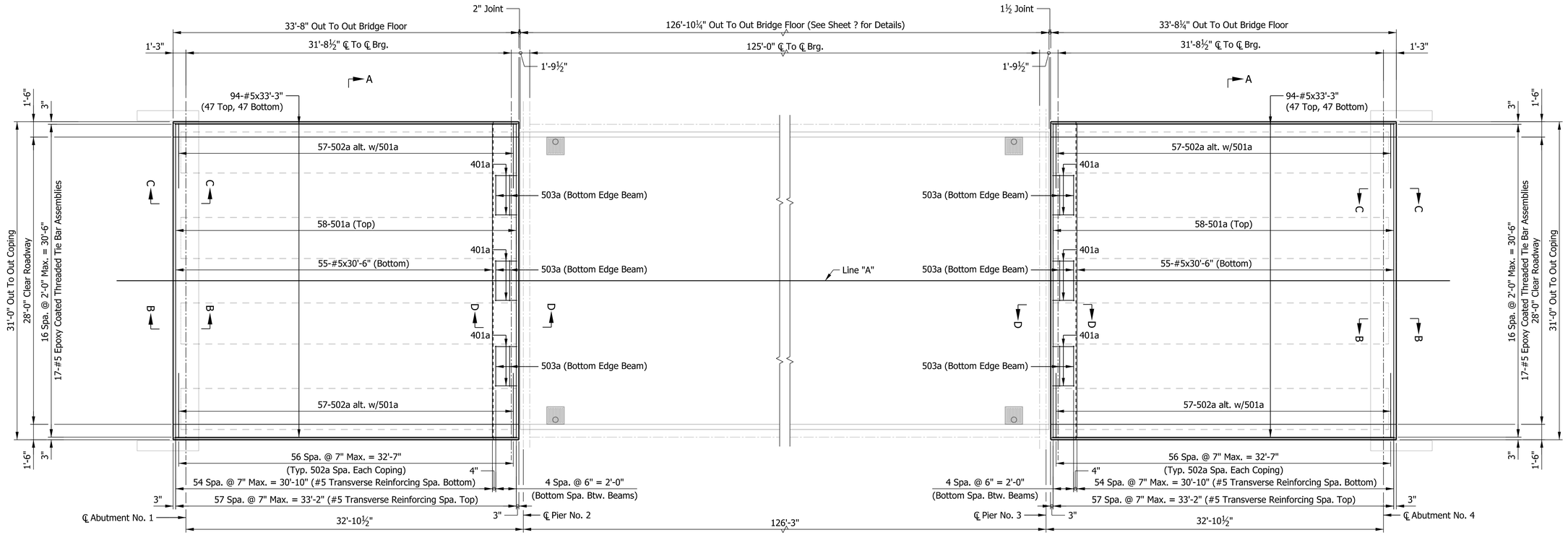
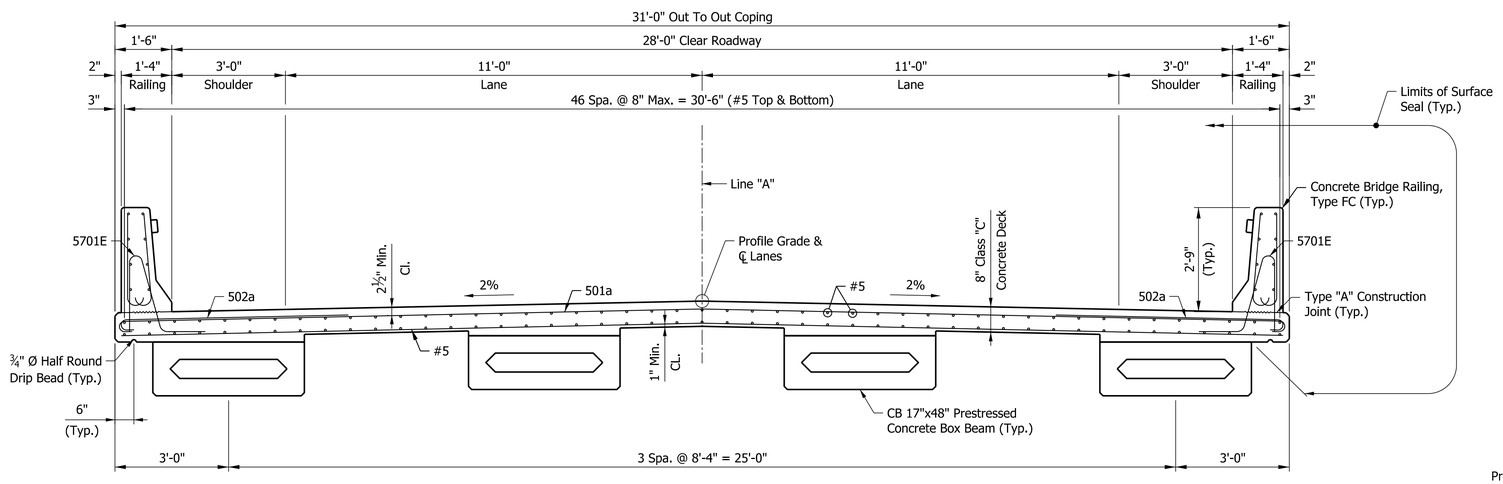


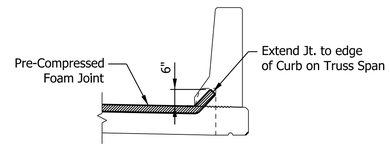
PLAN

Scale: 3/16" = 1'-0"



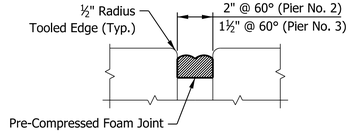
SECTION A-A

Scale: ½" = 1'-0"



JOINT DETAIL AT RAILING

No Scale



Pre-Compressed Foam Joint

No Scale

NOTES

1. All Reinforcing Steel to be Epoxy Coated.
2. For Bar Bending Diagrams and Bill of Materials, see sheet 25.
3. For Reinforcing Bar Notes and Standard End Hooks, see Standard Drawing E 703-BRST-01.
4. The top reinforcing in the deck shall be securely tied down to the deck forms and/or beams to prevent lifting during concrete placement.
5. Screed data will be furnished upon request.
6. Suitable restraint shall be provided to prevent the rotation of the beams, particularly the outside beam, from construction loads, such as the weight of the concrete deck, finishing machine, forms, etc.
7. For Sections B-B and C-C see Sheet No. 24.
8. For Sections D-D see Sheet No. 25.
9. For section thru railing and additional railing details, see Sheet No. 27.
10. For Edge Beam reinforcement details, see Sheet No. 25.
11. 5701E bar extending into the railing shall be placed and cast in deck, but billed with railings.
12. See Unique special provisions for material and installation of Pre-Compressed Foam Joint.
13. Expansion Length Pier No. 2:
    Span 1 = 16'-0"
    Span 2 = 125'-0"

    Expansion Length Pier No. 3:
    Span 2 = 0'-0"
    Span 3 = 16'-0"

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INDIANA
DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
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CONTRACT	PROJECT
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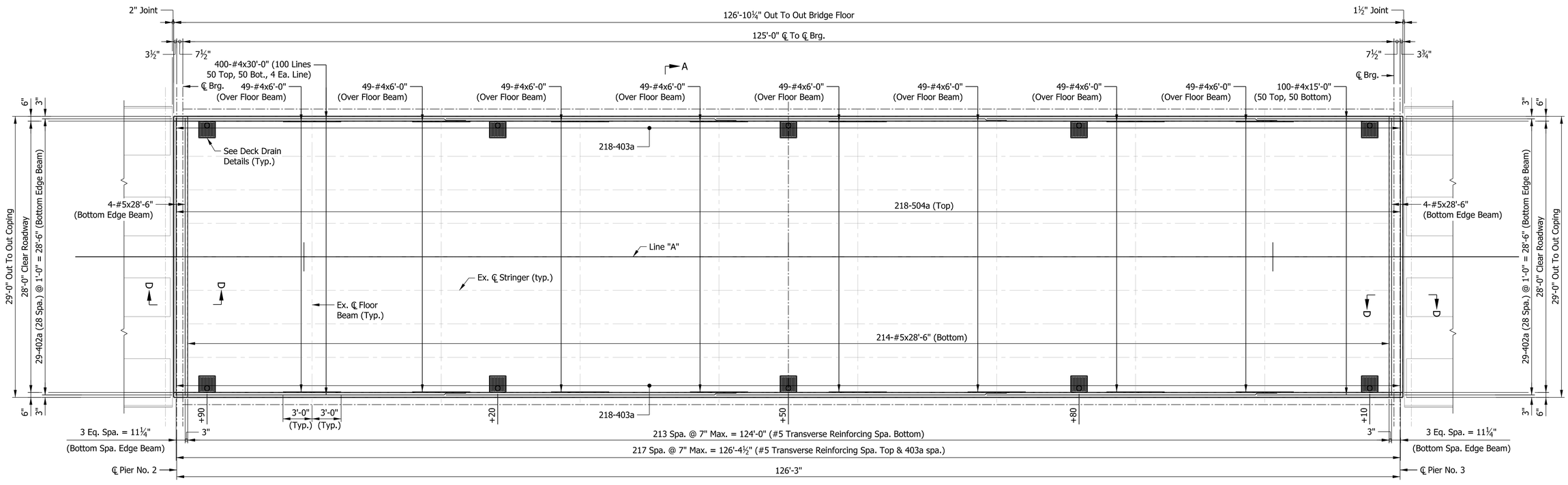
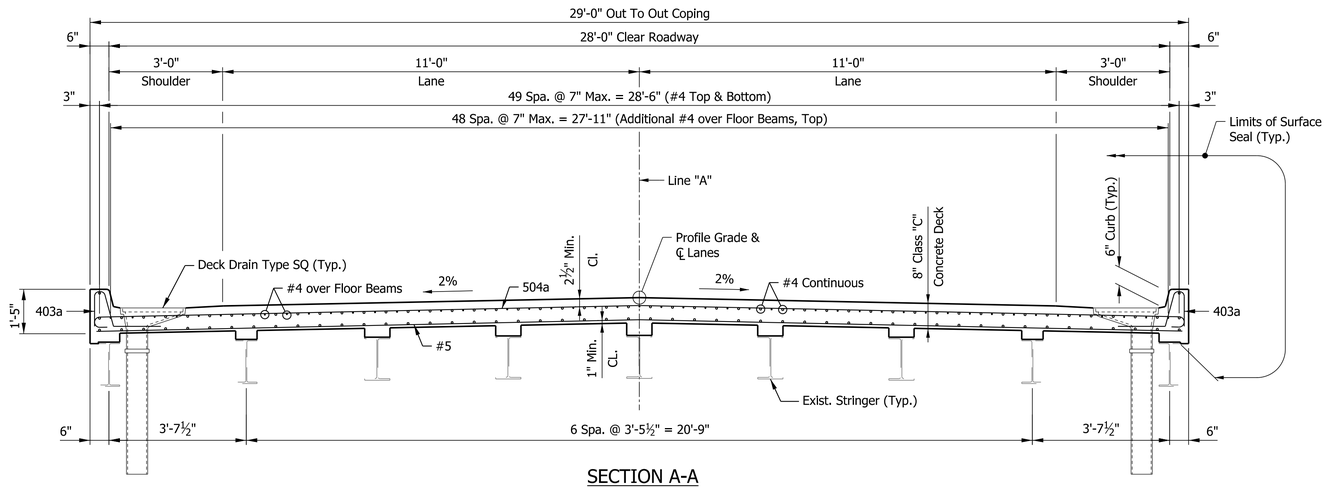


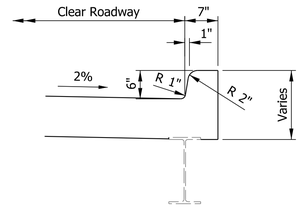
PLAN

Scale: 3/16" = 1'-0"



SECTION A-A

Scale: ½" = 1'-0"



CURB DETAIL

No Scale

NOTES

1. All Reinforcing Steel to be Epoxy Coated.
2. For Bar Bending Diagrams and Bill of Materials, see sheet 25.
3. For Reinforcing Bar Notes and Standard End Hooks, see Standard Drawing E 703-BRST-01.
4. The top reinforcing in the deck shall be securely tied down to the deck forms and/or beams to prevent lifting during concrete placement.
5. Screed data will be furnished upon request.
6. Suitable restraint shall be provided to prevent the rotation of the beams, particularly the outside beam, from construction loads, such as the weight of the concrete deck, finishing machine, forms, etc.
7. For Sections D-D see Sheet No. 25.
8. For deck drainage and collection system details, see sheet 26.

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CHECKED: KMP | CHECKED: KMP

INDIANA DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
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CONTRACT	PROJECT
B-42017	1593276

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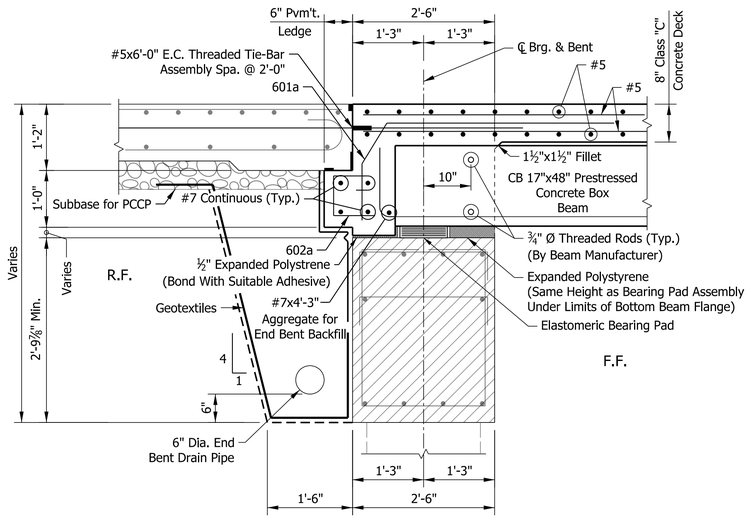
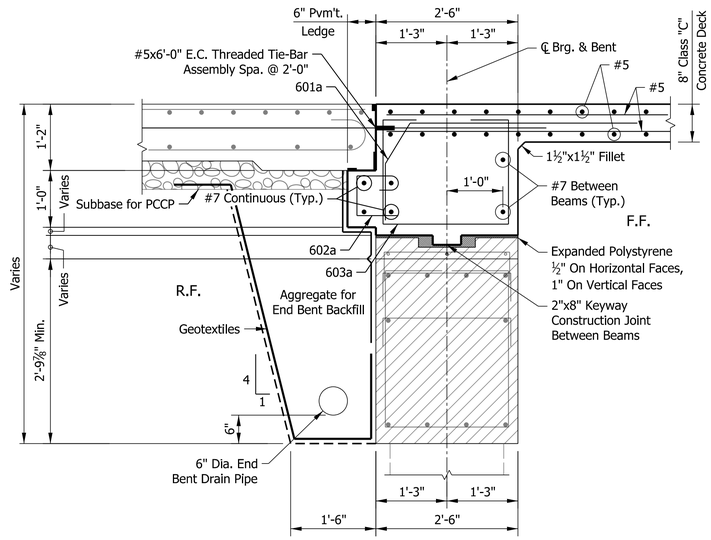


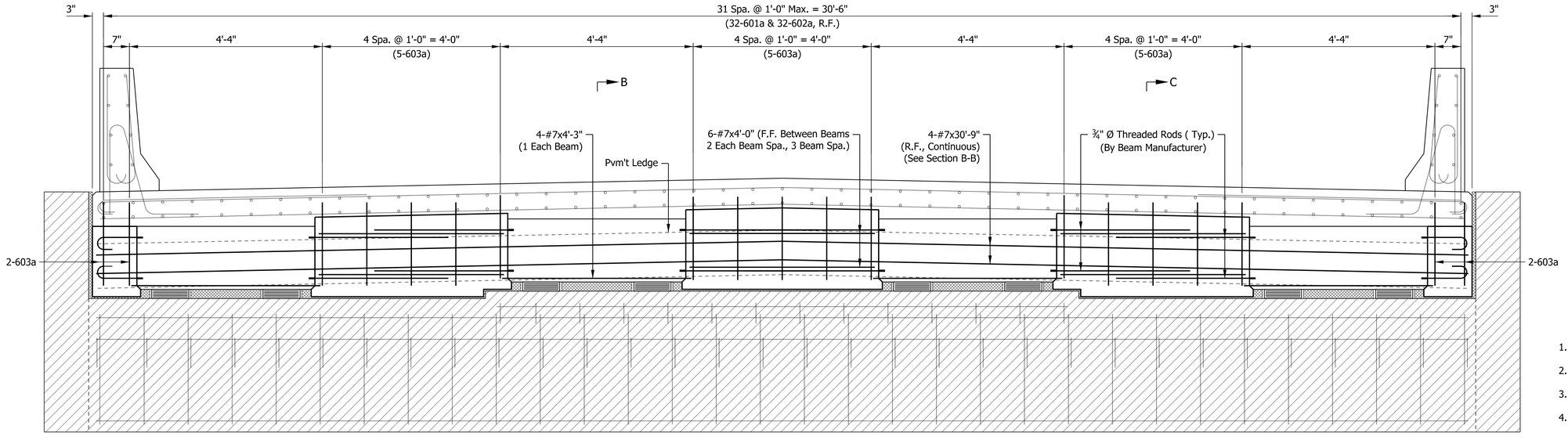
SECTION B-B

Scale: ¾" = 1'-0"



SECTION C-C

Scale: ¾" = 1'-0"



TYPICAL BENT DIAPHRAGM

Scale: ¾" = 1'-0"

NOTES

1. For Reinforcing Bar Notes, see Standard Drawing E703-BRST-01.
2. All reinforcing steel to be epoxy coated.
3. Screed data will be furnished upon request.
4. Suitable restraint shall be provided to prevent the rotation of the beams, particularly the outside beam, from construction loads, such as the weight of the concrete deck, finishing machine, forms, etc.
5. Hatched area to be poured with substructure.
6. For additional details and Bill Of Materials, see Sheet No. 25.
7. For section thru railing and additional railing details, see Sheet No. 27.

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SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
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	SHEETS
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CONTRACT	PROJECT
B-42017	1593276

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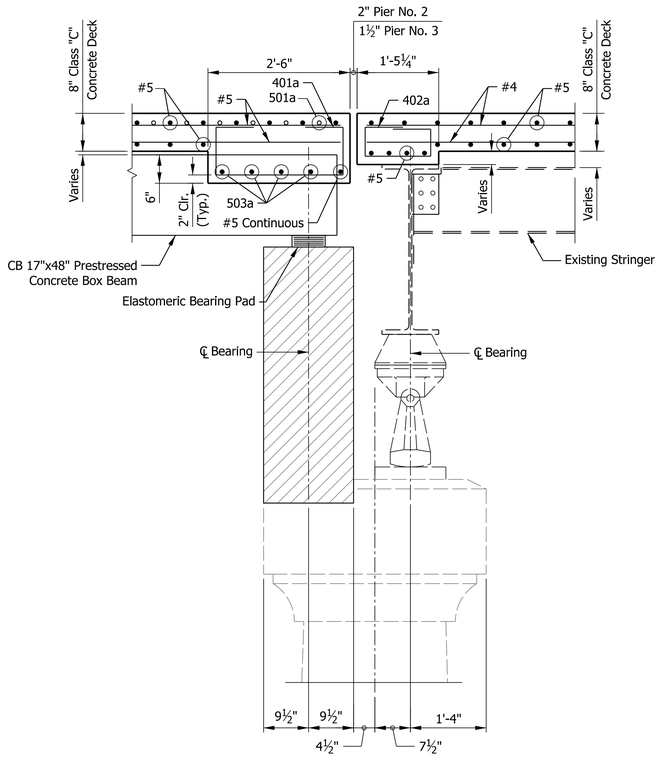
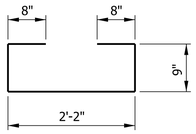
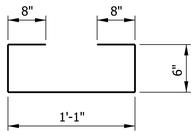


SECTION D-D

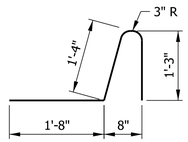
Scale: ¾" = 1'-0"



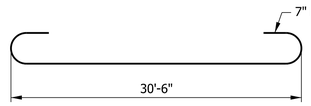
401a x 5'-0"



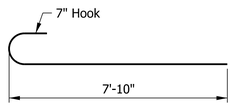
402a x 3'-5"



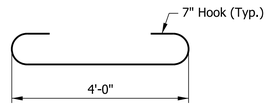
403a x 4'-4"



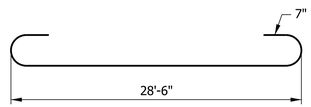
501a x 31'-8"



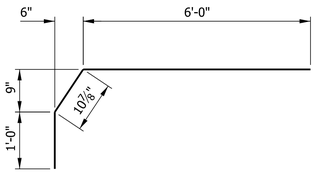
502a x 8'-5"



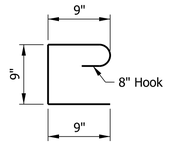
503a x 5'-2"



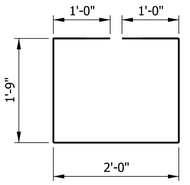
504a x 29'-8"



601a x 7'-11"

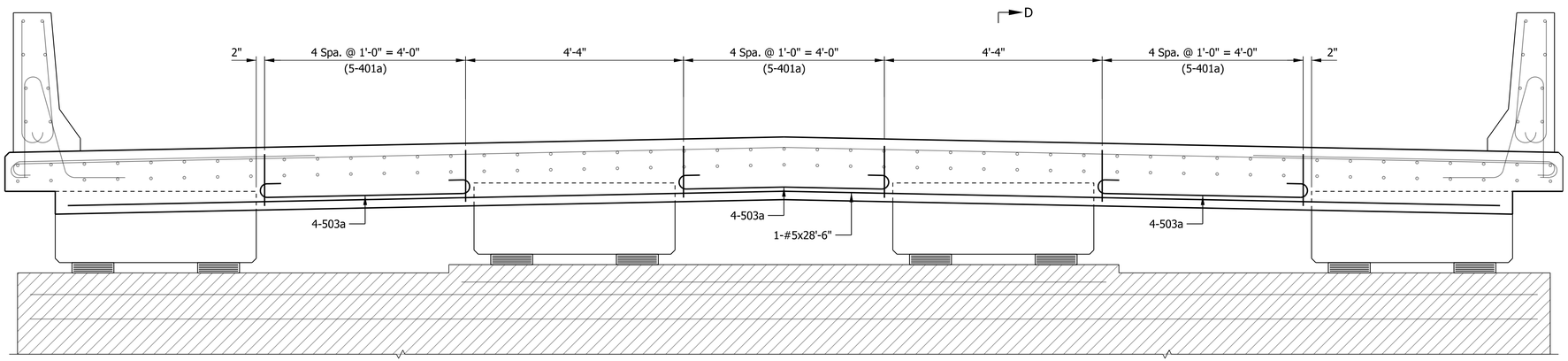


602a x 2'-11"



603a x 7'-6"

BILL OF MATERIALS SUPERSTRUCTURE			
PLAIN REINFORCING STEEL			
Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
#7	8	30'-9"	
#7	8	4'-3"	
#7	12	4'-0"	
TOTAL #7 BARS			670
601a	64	7'-11"	
602a	64	2'-11"	
603a	38	7'-6"	
TOTAL #6 BARS			1,469
501a	116	31'-8"	
502a	228	8'-5"	
503a	24	5'-2"	
504a	218	29'-8"	
#5	188	33'-3"	
#5	110	30'-6"	
#5	224	28'-6"	
TOTAL #5 BARS			29,385
401a	30	5'-0"	
402a	58	3'-5"	
403a	436	4'-4"	
#4	400	30'-0"	
#4	100	15'-0"	
#4	392	6'-0"	
TOTAL #4 BARS			12,083
TOTAL EPOXY COATED REINF.			41,468
CONCRETE			
CLASS "C" IN SUPERSTRUCTURE			175.4 CYS
MISCELLANEOUS			
SURFACE SEAL			6390 SFT
PRE-COMPRESSED FOAM JOINT			62 LFT
X-STD. ROADWAY DRAIN TYPE SQ WITH GRATE A 192 LBS. EACH			10 EACH
E.C. THREADED TIE-BAR ASSEMBLIES			34 EACH



TYPICAL CONCRETE EDGE BEAM

Scale: ¾" = 1'-0"

NOTES

1. For Reinforcing Bar Notes, see Standard Drawing E703-BRST-01.
2. All reinforcing steel to be epoxy coated.
3. Screed data will be furnished upon request.
4. Suitable restraint shall be provided to prevent the rotation of the beams, particularly the outside beam, from construction loads, such as the weight of the concrete deck, finishing machine, forms, etc.
5. Hatched area to be poured with substructure.
6. For section thru railing and additional railing details, see Sheet No. 27.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: JAS DRAWN: TMT

CHECKED: TDJ CHECKED: KMP

INDIANA DEPARTMENT OF TRANSPORTATION

SUPERSTRUCTURE DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
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CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:32:21 AM By: thomatm Pen: Transportation.tbl

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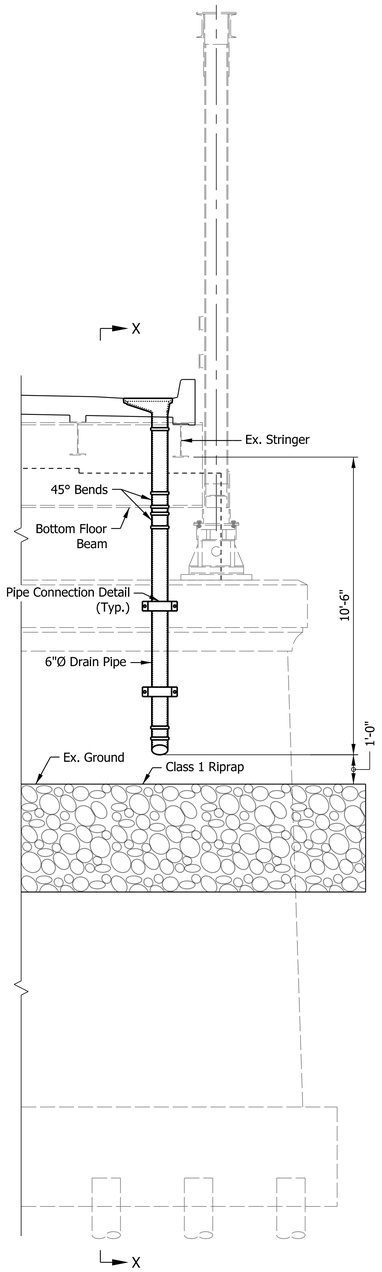
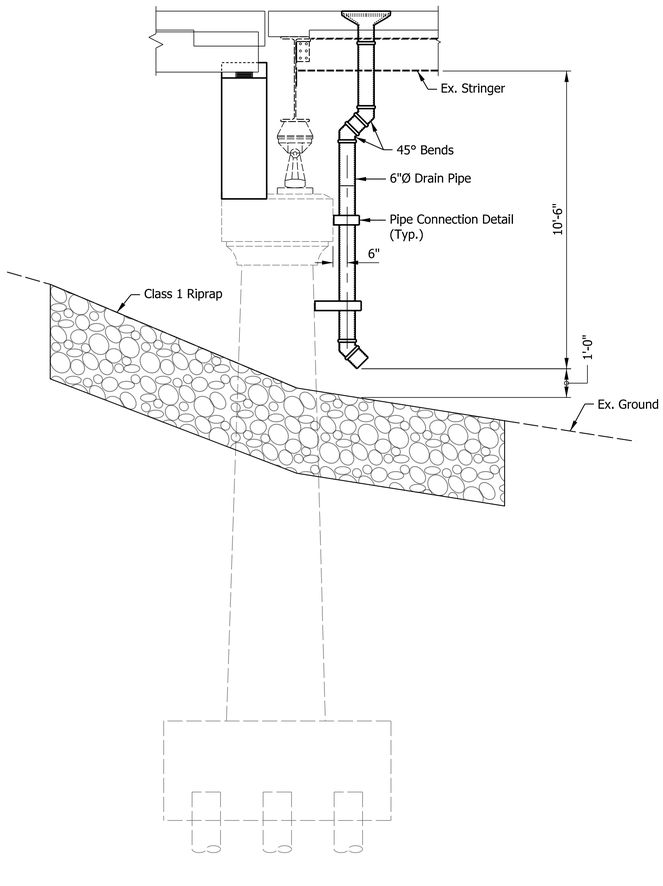


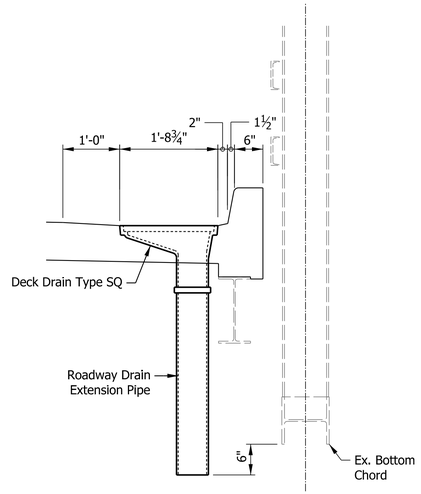
ELEVATION - PIER NO. 2 & 3

Scale: ⅜" = 1'-0"



SECTION X-X

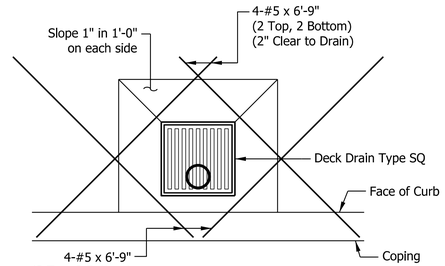
Scale: ⅜" = 1'-0"



Drain Section

Note

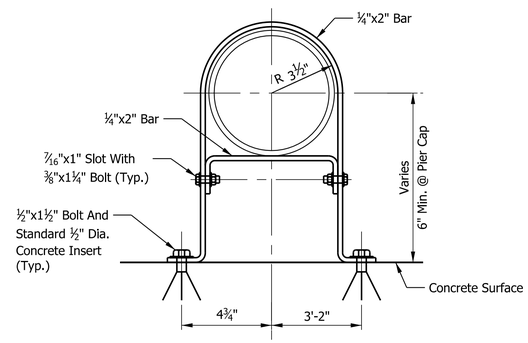
Cut and bend deck reinforcing as necessary to allow for drain installation.



Plan View

DECK DRAIN DETAILS

Scale: ¾" = 1'-0"



PIPE CONNECTION DETAIL

No Scale

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: NRT | DRAWN: TMT

CHECKED: KMP | CHECKED: KMP

INDIANA
DEPARTMENT OF TRANSPORTATION

DRAINAGE DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	26 of 30
CONTRACT	PROJECT
B-42017	1593276

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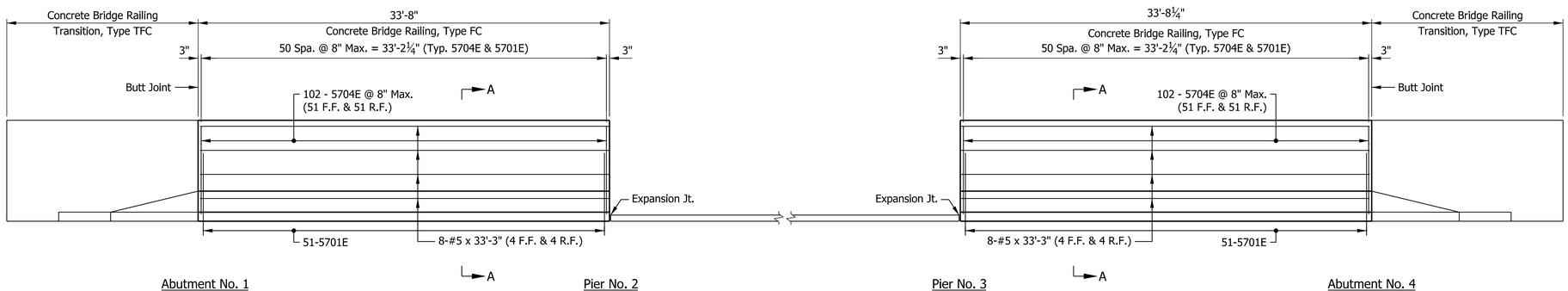


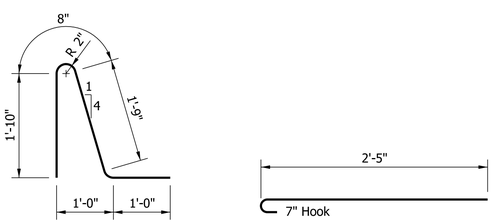
Abutment No. 1 | Pier No. 2 | Pier No. 3 | Abutment No. 4

CONCRETE RAILING ELEVATION

South Coping Shown, North Coping Same
No Scale

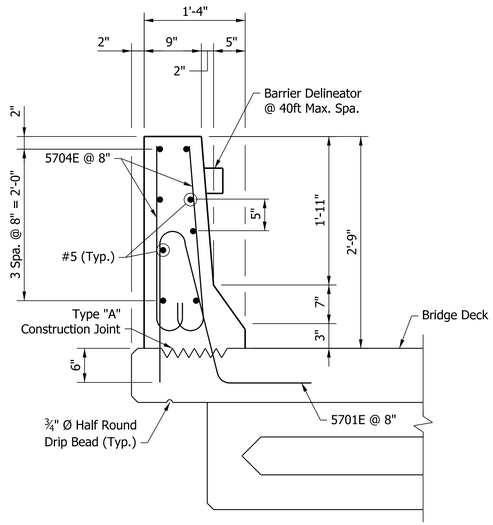
BILL OF MATERIALS CONCRETE BRIDGE RAILING			
EPOXY COATED REINFORCING STEEL			
Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
5701E	102	5'-6"	
5704E	204	3'-0"	
#5	16	33'-3"	
TOTAL #5 BARS			1,778
CONCRETE BRIDGE RAILING TRANSITION, TYPE TFC (2 @ 551			1102
TOTAL EPOXY COATED REINF.			2,880
CONCRETE			
RAILING, CONCRETE FC			68 LFT
CONCRETE BRIDGE RAILING TRANSITION, TYPE TFC			2 EACH
MISCELLANEOUS			
SURFACE SEAL			700 SFT
BARRIER DELINEATORS			6 EACH

2 REQUIRED



5701E x 5'-6"

5704E x 3'-0"



SECTION A-A
Scale: 1" = 1'-0"

NOTES

1. For Bridge Railing, Type FC details, see Std. Dwg. E 706-BRSF-02.
2. For TFC Transition details, see Std. Dwg. E 706-TTFC-01 through E 706-TTFC-03.
3. All reinforcing bars shall be Epoxy Coated.

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RECOMMENDED FOR APPROVAL _____ - DESIGN ENGINEER DATE

DESIGNED: TMT | DRAWN: TMT

CHECKED: KMP | CHECKED: JAS

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DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	27 of 30
CONTRACT	PROJECT
B-42017	1593276

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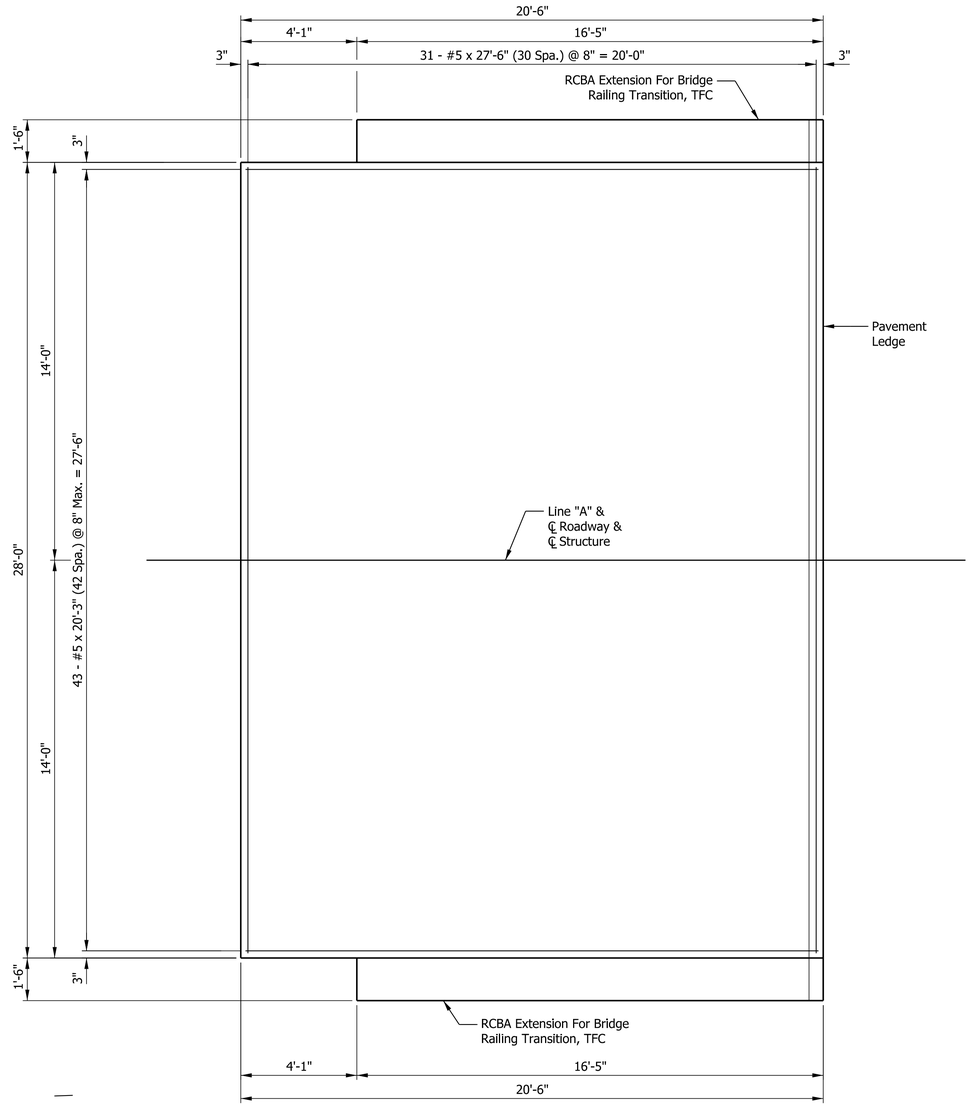
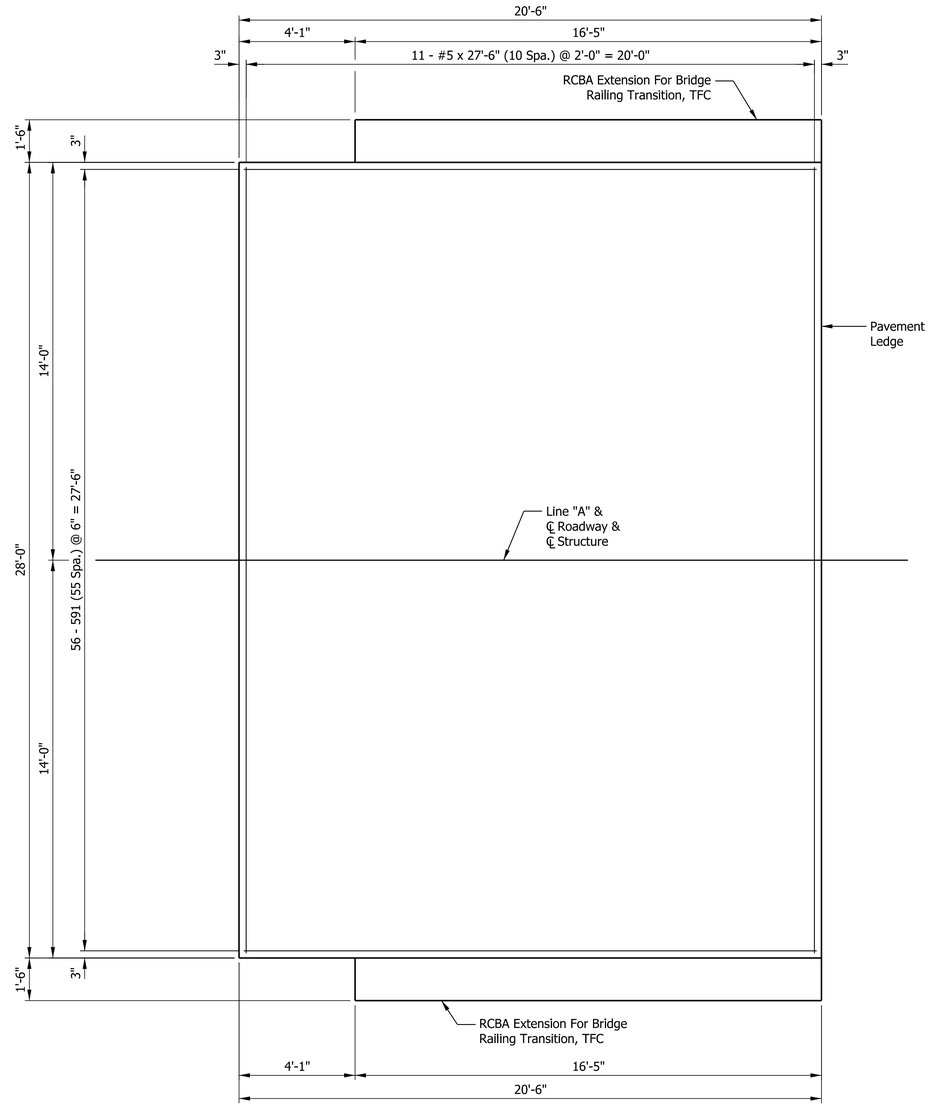


PLAN @ ABUTMENT NO. 1
Showing Top Reinforcement
Plan @ Abutment No. 4 Same by 180° Rotation
Scale: ⅜" = 1'-0"



PLAN @ ABUTMENT NO. 1
Showing Bottom Reinforcement
Plan @ Abutment No. 4 Same by 180° Rotation
Scale: ⅜" = 1'-0"

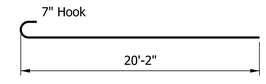
BILL OF MATERIALS
R.C. BR. APPR. BENT NO. 1

EPOXY COATED REINFORCING STEEL			
Size & Mark	Number of Bars	Length (Ft. - In.)	Weight (Lbs.)
591	56	20'-2"	
#5	42	27'-6"	
#5	43	20'-3"	
TOTAL #5 BARS (RCBA)			3,291
CONCRETE BRIDGE APPROACH EXTENSION (2 @ 269)			538
CONCRETE BRIDGE RAILING TRANSITION, TYPE TFC (2 @ 551)			1102
TOTAL EPOXY COATED REINF.			4,931

CONCRETE	
R.C. BRIDGE APPROACH, 12 IN	69.3 SYS
CONC. BR. RAILING TRANS., TYPE TFC	2 EACH

MISCELLANEOUS	
SURFACE SEAL	595 SFT
SUBBASE FOR PCCP	18 CYS
GEOTEXTILE FOR SUBGRADE TYPE 2B	69.3 SYS

R.C. BR. APPR. BENT NO. 4 SAME



591 x 20'-9"

NOTES:

1. For reinforcing bar notes, see Standard Drawing E 703-BRST-01.
2. For TFC Transition details, see Standard Drawing E 706-TTFC-01 through E 706-TTFC-03.
3. For RCBA Extension for Bridge Railing Transition, Type TFC. See Standard Drawing E 609-TBAE-01.
4. For Construction Joint Type I-A see Standard Drawing E 609-BRJT-01.
5. All reinforcing bars shall be Epoxy-Coated.
6. RCBA's and TFC Transitions shall be surface sealed.
7. For section through approach and additional details, see Standard Drawings E 609-RCBA-01 thru E 609-RCBA-03.

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RECOMMENDED FOR APPROVAL _____ DESIGN ENGINEER _____ DATE

DESIGNED: TMT | DRAWN: TMT

CHECKED: KMP | CHECKED: JAS

INDIANA
DEPARTMENT OF TRANSPORTATION

APPROACH SLAB DETAILS

SCALE	BRIDGE FILE
AS NOTED	(421)39-12-01792 C
	DESIGNATION
	1593276
	SHEETS
	28 of 30
CONTRACT	PROJECT
B-42017	1593276

Plot: 2/14/2020 10:32:33 AM By: thomatm Pen: Transportation.tbl

SUMMARY OF BRIDGE QUANTITIES

ITEM	CONCRETE			BARRIER, DELINEATOR	RAILING, CONCRETE, FC	CONCRETE BRIDGE RAILING TRANSITION, TFC	REINFORCEMENT			REINF. CONCRETE BRIDGE APPROACH 12 IN.	SUBBASE FOR PCCP	GEOTEXTILE FOR SUBGRADE TYPE 2B	EXCAVATION, FOUNDATION, UNCLASSIFIED	PRECOMPRESSED FOAM JOIN	PATCHING CONCRETE STRUCTURES	EMBEDDED GALCANIC ANODE	FIELD DRILLED HOLES IN CONCRETE	DRILL HOLES	AGGREGATE FOR END BENT BACKFILL	GEOTEXTILE FOR UNDERDRAIN TYPE 3	PIPE, END BENT DRAIN, 6 IN	X-STD. ROADWAY DRAIN TYPE SQ WITH GRATE A 192 LBS. EACH	STRUCTURAL STEEL *	STRUCTURAL STEEL**	CONCRETE BOX BEAM 17 IN. X 48 IN.	SURFACE SEAL*
	CLASS C SUPERSTR.	CLASS A SUBSTR.	CLASS C SUBSTR.				REINF. BARS, EPOXY COATED	REINF. BARS	THREADED TIE BAR ASSEMBLY, EPOXY COATED																	
	CYS	CYS	CYS	EACH	LFT	EACH	LBS	LBS	EACH	SYS	CYS	SYS	CYS	SFT	SFT	EACH	EACH	EACH	CYS	SYS	LFT	EACH	LBS	LBS	LFT	SFT
SUPERSTRUCTURE	175.4						41468		34					62				28				10	1719	1000	262	6390
BENT No. 1			10.7				2103						10						10		51					
PIER No. 2		7.9					739								100	39	31			30						255
PIER No. 3		7.8					739								100	39	31			30						255
BENT No. 4			10.5				2103						10						10		51					
APPROACHES							9862			138.6	36	138.6														1190
CONCRETE BRIDGE RAILING				8	135	4	5760																			1400
TOTALS	175.4	15.7	21.2	8	135	4	62774	0	34	138.6	36	138.6	20	62	200	78	62	28	20	60	102	10	1719	1000	262	9490

* Estimated quantity, to be paid for as "LSUM"

** Estimated quantity, work to be constructed at the direction of the Engineer, see Special Provisions

STRUCTURAL STEEL BILL OF MATERIALS

Member	Member Piece	Locations L0-X	L0-Y	L9-X	L9-Y	L7U8-Y	Total No.	Width (in)	Thickness (in)	Height (in)	Weight (lb)	Total Weight (lb)
Gusset Plate	Gus PL 39 1/4" x 7/16" x 3' 3 3/4"	2	2	2	2		8	39.25	0.4375	39.75	193.6	1548
Cover Plate	PL 17" x 3/8" x 1' 8"	1	1	1	1		4	17	0.375	20	36.2	145
Cover Plate	PL 8" x 3/8" x 2' 7"					1	1	8	0.375	31	26.4	26
	Total											1719

RIVET REMOVAL SUMMARY

Location	Member Piece	No.
L0-X (SW)	Outside Gusset Plate	75
	Inside Gusset Plate	75
	L0 U1-X Cover Plate	18
	Top Cover Plate	28
	End Post	4
	Subtotal	200
L0-Y (NE)	Outside Gusset Plate	75
	Inside Gusset Plate	75
	L0 U1-Y Cover Plate	18
	Top Cover Plate	28
	End Post	4
	Subtotal	200
L9-X (NW)	Outside Gusset Plate	75
	Inside Gusset Plate	75
	U8 L9-X Cover Plate	18
	Top Cover Plate	28
	End Post	4
	Subtotal	200
L9-Y (SE)	Outside Gusset Plate	75
	Inside Gusset Plate	75
	U8 L9-Y Cover Plate	18
	Top Cover Plate	28
	End Post	4
	Subtotal	200
	Total	800

BOLT REMOVAL SUMMARY

Location	Member Piece	No.
L0-X (SW)	Top Cover Plate	10
L0-Y (NE)	Top Cover Plate	10
L9-X (NW)	Top Cover Plate	10
L9-Y (SE)	Top Cover Plate	10
	Total	40

DRILL HOLE SUMMARY

Location	Member Piece	No.
L0-X (SW)	L0 L1 L2-X	4
L0-Y (NE)	L0 L1 L2-Y	4
L9-X (NW)	L7 L8 L9-X	4
L9-Y (SE)	L7 L8 L9-Y	4
L7 U8-Y	L7 U8-Y	12
	Total	28

PAVEMENT QUANTITIES AND APPROACH TABLE

LOCATION	DESCRIPTION (APPROACH TYPE OR CLASS)	CURBS		PAVEMENT REMOVAL	EXCAVATION, COMMON		QC/QA-HMA, 3, 64			ASPHALT MATERIAL FOR		COMPACTED AGGREGATE, NO.53		MILLING, TRANSITION	SUBGRADE TREATMENT, TYPE IC	LIQUID ASPHALT SEALANT	JOINT ADHESIVE, SURFACE	JOINT ADHESIVE, INTERMEDIATE	REMARKS
		CURB, ASPHALT, REMOVE	CURB HMA				SURFACE 9.5 mm	INTERM 19.0 mm	BASE 25.0 mm	PRIME COAT	TACK COAT	DEPTH							
							LBS PER SYD												
							165	275	1210			6"	3"						
		LFT	LFT	SYS	CYS	CYS	TONS	TONS	TONS	SYS	SYS	TONS	TONS	SYS	SYS	LFT	LFT	LFT	
NORTH APPROACH				50	85		36				373.5	43		373.5		360	360		
SOUTH APPROACH				50	85		31				373.5	43		373.5		360	360		
TOTALS		0	0	100	170		67	0	0		747	86	0	747	0	720	720	0	

SHEET SIGN & POST SUMMARY

SIGN									POST								
PLAN SHEET NO. / LINE	SIGN EXISTING LOCATION (STA.)	SIGN PROPOSED LOCATION (STA.)	SIGN CODE	SIGN SIZE (IN x IN)	GROUND - MOUNTED SIGN AREA (ft^2)			MOUNTED ON SIGN CATENARY, AREA (ft^2)	U CHANNEL				SQUARE				
													2" X 2" - 12 GA. (TYPE 2)			2 1/4" X 2 1/4" - 12 GA. (TYPE 1)	
									POST LENGTH		POST LENGTH		REINFORCED ANCHOR			REINFORCED ANCHOR	
									1	2	TYPE "A"	TYPE "B"	POST LENGTH (FT.)			POST LENGTH (FT.)	
					0.080"	0.100"	0.125"	0.080"	FT.	FT.	FT.	FT.	1	2	TOTAL	1	TOTAL
Line "A"																	
5	41+65	41+65	Wildcat Creek													5.5	5.5
	42+34	42+15	Mile Post													5.5	5.5
	42+45	42+24	End Marker													5.5	5.5
	42+45	42+24	End Marker													5.5	5.5
	44+55	44+80	End Marker													5.5	5.5
	44+55	44+80	End Marker													5.5	5.5
					0.00	0.00	0.00	0.00							0		33.0

GUARDRAIL SUMMARY TABLE

LOCATION			W-BEAM		TRANSITIONS		END TREATMENTS			
APPROACH	LEFT	RIGHT	MGS GUARDRAIL, W-BEAM, 6 FT 3 IN. SPACING	GUARDRAIL, W-BEAM, 6 FT 3 IN. SPACING	GUARDRAIL, TRANSITION, MGS WITHOUT CURB	GUARDRAIL, TRANSITION, TYPE TGB	GUADRAIL END TREATMENT, OS	GUARDRAIL, TERMINAL SYSTEM, W-BEAM CURVED, 1	GUARDRAIL, END TREATMENT, REMOVE	GUADRAIL, REMOVE
			LFT	LFT	EACH	EACH	EACH	EACH	EACH	LFT
NORTH APPORACH		X		62.5		1		1		120
NORTH APPORACH	X		87.5		1		1			140
SOUTH APPROACH		X	81.25		1		1			140
SOUTH APPROACH	X		37.5		1		1			140
TOTALS			206.25	62.5	3	1	3	1	0	540

RIPRAP, SEEDING, AND SODDING SUMMARY TABLE

LOCATION										RIPRAP TURNOUT		RIPRAP FOR SCOUR			SODDING		WATER	SEEDING					
FROM STATION	TO STATION	LEFT	RIGHT	BENT 1	PIER 2	PIER 3	BENT 4	LEFT	RIGHT	RIPRAP, REVETMENT	GEOTEXTILE FOR RIPRAP, TYPE 3	RIPRAP, REVETMENT	RIPRAP, REVETMENT	GEOTEXTILE FOR RIPRAP, TYPE 3	FOR PAVED SIDE DITCHES	FOR TURNOUTS		MOBILIZATION AND DEMOBILIZATION FOR SEEDING	MULCHED SEEDING U	MULCHED SEEDING R	MULCHING MATERIAL	FERTILIZER
										TONS	SYS	TONS	TONS	SYS	SYS	SYS	KGAL	EACH	SYS	SYS	TON	TON
			X							25	40					15	1					
		X								25	40					15						
									X	25	40					15						
								X		25	40					15						
				X								225		300								
					X								106	85								
						X							106	85								
							X					225		300								
TOTALS										100	160	450	212	770		60	1	1		1000		